

From cells to councils: a conservation law of allocation under scarcity

James Miller

Independent researcher

contact@n-ought.com

ORCID: 0009-0009-6595-647X

Version 2, October 2026

Record: <https://doi.org/10.5281/zenodo.23222615>

Abstract

Under scarcity, allocation is zero-sum. Outside a few exceptions, named in advance, no ordering of access in a system short of a resource reduces its total shortfall: it only decides which parts go short, and correcting a deficit at one part leaves an equal deficit elsewhere. We state this as a conservation law, derived from a general model of how systems that must persist ration resources: all parts draw on one shared flow of each resource, each part's access is limited separately, and a governor protects the flow to one part, the top, by limiting everyone else's access first. Its ranked order of draws on a shared flow has been found separately in at least four fields, modelled or measured: organism energetics, human metabolism (the Selfish Brain), cell bioenergetics and plant carbon allocation. We found none carried beyond its domain. We reached it from principles about institutions and state the full architecture once, in seven components. From the general form follow results that none of the theories compared in Table 3 states formally across domains, including why a steady protected flow carries no information about the shortfall beneath it, the reserve at failure, when the speed of a decline causes lasting harm, and when rerouting overloads a route despite spare capacity. Of two pre-registered, blind-adjudicated tests, a predicted early warning before falls in arterial pressure in surgical cases with heavy blood loss failed, at half weight; this version finds errors in the prediction's derivation and in the test's mapping, and corrects both; whether they caused the failure is untested, and the verdict stands. The prediction that the order of loss follows access documented in advance was supported, at reduced weight, in English council spending, though not the version tied to deeper funding cuts. We state what would refute the model.

1. Introduction

Systems that persist under finite resources show a recurring pattern. When resources run short, the flow to the part that matters most is protected, while parts that matter less are run down out of sight. In a bleeding patient the protected flow is the oxygen the brain draws from its blood supply, measured against its resting need; in models calibrated to simulated and experimental blood loss, the vessels of gut and kidney narrow (Table 4), while arterial pressure, the figure clinicians watch, is held

near normal as cardiac output falls (Evans et al. 2001). Under energy restriction the brain's own energy supply has priority (Peters et al. 2004), and brain mass, which observers can weigh, stays almost unchanged while body mass falls (Sprengell, Kubera and Peters 2021a). In English councils between 2014-15 and 2019-20, spending on lines with a statutory duty grew faster than on discretionary lines (Section 5.1), though the gap was not detectably wider where funding fell more. The balanced budget that observers watch is a level the council's rules hold. In a tree in drought the water-conducting tissue fails while canopy colour, which observers watch, lags the failure (Hammond et al. 2019); the tree's top, and so its protected flow, would be fixed by a mapping made in advance (Section 3.1). In the patient, the animal and the tree, the figure observers watch said little about what was happening beneath it until late. That is an observation of these cases. Proposition 2 concerns the protected flow, and whether a watched figure tracks it is a mapping question (Section 3.1).

The law. Under scarcity, allocation is zero-sum. At a fixed, fully used inflow, raising one part's access lowers another's, and while no part receives more than it requires, no setting of access reduces the total shortfall: it only decides which parts go short. The total falls only if resource is added, what parts require is lowered, a gate that holds resource back is opened, or what a part receives beyond what it requires is taken back. Every unit of the gap is met from a store, met from outside the boundary, or left unmet at a named part. Drawing a store meets the gap now and leaves less for later; drawing from outside meets it from beyond the boundary.

As algebra, this is an accounting identity (Proposition 1). Kleinrock's (1965) conservation law for queues is one too: in a work-conserving single-server queue, no non-preemptive priority discipline changes the load-weighted sum of waiting times across classes; it only moves delay between them. Kleinrock's order is a position in a queue; the model's order is priority set by access. In Kleinrock's law delay moves between classes; here nothing moves: the total shortfall is fixed, and access decides only which parts go short. The content of such a law is not its algebra. It lies in three places:

- **the reference:** each part's requirement is fixed in advance and never lowered because the part has shut down, so a shortfall cannot be defined away;
- **the exceptions:** they are named and can be checked;
- **practice:** how routinely it acts as if the law were false, by treating the visible figure.

Four received views the law contradicts under scarcity.

- **"Correcting an abnormal reading treats the problem."** In a resource-short system, correcting one part's shortfall outside the named exceptions only changes which parts go short (Proposition 1).
- **"A stable reading means a stable system."** The protected flow is held by drawing stores and outside input and by lower-ranked parts going short. While these cover the gap it carries no information about the shortfall beneath it, and it breaks only when they no longer can, which under proportional release leaves part of the store unused (Propositions 2 and 3). A reading the governor holds is kept steady by the same spending while its levers last; we state this in words only, not as a proposition.
- **"Resilience is a property a system has."** Under scarcity, resilience is a store being drawn, input from outside, or a lower-ranked part going without (Proposition 1).

- **"The failing organ is the problem."** Only sometimes, and the model says when. If parts outside the top and its supports are still supplied when the top or a support part fails, the supports were met in the step before, and no unit of the top or a support part is still coming back from an earlier shortfall, then with true signals the cause is a cut or narrowed supply line, or damage to the part (S1.9); a signal fault is the other candidate (S9.5). If lower-ranked parts have already gone short, the failure is the last visible step of a system-level allocation: look down the order (Proposition 4), and at where the shortfall began, which may be falling supply or rising requirement, including the part's own after damage (Section 3.4).

What is new. The same ranked order of draws on a shared flow has been found at least four times, modelled or measured. Organism energetics (Kooijman 2010), the Selfish Brain account of human metabolism (Peters et al. 2004), the hierarchy of ATP consumers in cells (Buttgereit and Brand 1995) and sink priority in plants (Minchin, Thorpe and Farrar 1993) each describe a ranked order of draws on a shared flow, with the lowest-ranked going short first. Each was built for one domain, and our searches found none carried into another. Where the order was measured (in cells), survived pre-registered review (the brain under caloric restriction) or emerged from mechanism in transport models rather than being assumed (in phloem), it is not a convenience of one model. We arrived at the same ranked order from a fifth direction, principles about conservation and priority in institutions, and state the full architecture once, as the Persistence Allocation Model. Three things none of them carries:

1. **an order of loss fixed in advance from documented properties of access** (pathway constriction, autoregulation, affinity, statutory status) **and the rules for requirement**, never read off the observed order;
2. **a conserved ledger of shortfall**, which counts unmet requirement against a fixed reference and records the part at which every unit of it goes unmet;
3. **application outside bodies.**

Of the results we derive from the general form, eight are stated formally across domains by none of the theories compared in Table 3.

Tests. We report two pre-registered held-out tests, each replicated and adjudicated blind. One failed, at half weight and with two qualifications; one gave partial support, in the model's first held-out test outside bodies (Section 5). This version finds errors in the prediction's derivation and in the test's mapping, and corrects both; whether they caused the failure is untested, and the verdict stands (Section 5.1).

This paper:

- Section 2: the four instances, and theories that state parts of the architecture elsewhere;
- Section 3: the model, in seven components, its reduced form, the aids used to map it, and four propositions, which assume the governor's signals are true (thirteen further results are in Supplement S1);
- Section 4: the predictions, and how they separate the model from its neighbours;
- Section 5: the tests;
- Section 6: scope, limits and failures;
- Section 7: what follows;
- Changes from version 1: every substantive change, and why.

2. Four instances of one ranked order

Organism energetics. Dynamic Energy Budget (DEB) theory is the instance for whole organisms (Kooijman 2010). It ranks the uses of reserve, paying somatic maintenance before growth and reproduction (the κ rule). It releases reserve in proportion to its content, shrinks structure when reserve cannot pay maintenance, and treats defence as "more facultative" than somatic maintenance. It conserves mass and energy. It ranks uses, not organs; it keeps no ledger of unmet requirement; and it allocates no repair, since damage in DEB is irreparable. We import three of its forms (Section 3). DEB's maintenance-first order holds within a part: what reaches a part covers its upkeep before its work. Across parts, rank comes first, because, where the governor sets a part's access, it sets it as a whole.

The brain. The Selfish Brain theory is the instance in human metabolism, with the brain as both the protected consumer and the regulator of the body's energy supply (Peters et al. 2004). The brain gives priority to its own supply by inhibiting glucose uptake into muscle and fat. In later formulations it suppresses insulin, closing the insulin-dependent route into muscle and fat while drawing through an insulin-independent one. This is set out as an energy-conserving supply-chain model (Peters and Langemann 2009), and a brain-centred compartment model has been analysed formally (Göbel and Langemann 2011). Peters, McEwen and Friston (2017) link it to the free energy principle: under uncertainty the brain demands extra energy from the body, and if it cannot reduce the uncertainty, a persistent cerebral energy crisis may develop that burdens the individual as allostatic load. The link between brain and body is stated in words. Its order has survived pre-registered systematic review: under caloric restriction the brain lost almost no mass while the body lost a great deal (Sprengell, Kubera and Peters 2021a). It is formal for two compartments and one resource. It has no order fixed in advance across many parts and no ledger.

The cell. In thymocytes the ranked order was measured. A hierarchy of ATP consumers loses supply in order: macromolecule synthesis first, ion pumping later, proton leak last (Buttgereit and Brand 1995). It is a measurement, without a general model. Here access is built into the parts, through their affinity for ATP, and is not set by a governor. Within a snapshot a built-in setting is fixed, as a governor's setting is, but a change in a part's requirement can move its place; rank under saturable uptake (S1.7) is the formal case.

Plants. In phloem transport the ranked order emerges from mechanism: "sink priority" follows from the transport network and the kinetics of the sinks (Minchin, Thorpe and Farrar 1993). Access is built into the network and the sinks and is not set by a governor; within a snapshot it is fixed, as a governor's setting is, but a change in a sink's requirement can move its place (S1.7). Hierarchical plant growth models instead assume strict priority among organs (Grossman and DeJong 1994; classified as hierarchical by Marcelis and Heuvelink 2007). The order is assumed in three places: hierarchical plant models; DEB, where maintenance priority is a postulate; and the Selfish Brain, where it is the theory's premise. It was measured in cells, emerges from mechanism in phloem, and has been tested in the brain.

Parts of the architecture stated elsewhere.

- The energetic model of allostatic load (Bobba-Alves, Juster and Picard 2022) proposes that the energetic cost of stress first uses up reserve capacity, then squeezes growth, maintenance and repair, potentially without raising total energy expenditure. That is the model's hidden shortfall, with growth, maintenance and repair squeezed, stated in words.

- Triage theory (Ames 2006) proposes that scarce micronutrients go to proteins needed for short-term survival over those needed for long-term health, partly through binding affinity, at the level of enzymes, cells and organs.
- A unifying theory of hypoxia tolerance (Hochachka et al. 1996) describes a balanced suppression of energy supply and demand: ion pumping and protein synthesis are cut back (channel and translational arrest) while the cell's energy state is held.
- Allostasis (Sterling 2012) treats regulation as predictive and brain-led.
- The central governor model of exercise (Noakes 2012) holds that exercise is stopped before any system fails, with a reserve of motor units always kept. It is contested: Shephard (2009) finds convincing experimental evidence for its corollaries lacking, and argues that a plateau in oxygen consumption at exhaustion counts against a limiting central governor.
- The selfish immune system (Straub 2014) adds the immune and repair system as a second claimant that can take control of the body's spare energy. In this model, the brain and immune regulators are two modes of one governor function. Straub's chronic case, the immune system as a standing claimant that changes the order, is the open question on modes (Section 6).
- Ashby (1952) defined adaptive behaviour as behaviour that keeps an organism's essential variables within physiological limits; when an essential variable moves far from its normal value, step-functions change the system's behaviour. It is an early form of the governor: targets on sensed levels, with no shared resource, store or ledger.
- Perceptual control theory (Powers 1973) describes hierarchies in which higher levels set the reference values of lower ones. It gives the governor its form, but has no resource, store or ledger.
- The free energy principle accounts for action, perception and learning as the minimising of surprise (Friston 2010). A heuristic proof, with simulations, suggests that any (ergodic) random dynamical system with a Markov blanket will appear to act on its world to preserve its integrity (Friston 2013). It supplies a partial form of the governor, a regulator of the system's own states, and the framing of persistence, with no shared resource, ranked access or ledger.

What the convergence shows. These fields developed in separate literatures; in what we read, none of the four cites another's allocation model. Their agreement is evidence that the ranked order is real. It is not evidence that the results derived from it here are true: those stand or fall by test (Sections 4 and 5).

Table 1 condenses the comparison; the full table, with twenty theories, is Supplement S3.

Table 1. Prior theories against the model's features (F, formal; W, stated in words; E, measured, without a general model; p, partial or different form; blank, not found in what we read; bold, no theory in the full comparison, Supplement S3, has it in formal form).

Theory	Governor	Order from access	Order fixed in advance	Stores drawn first	Repair competes for access	Shortfall ledger	Outside bodies
Dynamic Energy Budget	p	p		F		p	
Selfish Brain	F	F (two compartments)	p	F		p	
Allostatic load (energetic)	W	W		W	W	W	
Triage theory		W	p	W	W	W	

Theory	Governor	Order from access	Order fixed in advance	Stores drawn first	Repair competes for access	Shortfall ledger	Outside bodies
Plant hierarchical and transport-resistance models		F		p		p	
Cell ATP hierarchy		E/F (measured)			E		
This model	F	F	F	F	F	F	F

3. The model in formal terms

In plain words. All parts draw on one shared flow of each resource. Each part's access to that flow is limited separately, by a setting the part does not control (a part that sets its own access, such as a tumour, is a separate case; Section 7). The governor protects the top by limiting everyone else's access first. Nothing passes between parts; a shortfall is a count at a part, not something that moves. No work or shortfall is handed from one part to another. Resource moves only through the shared flow and its routes, and one part's state changes another's only through a dependency or through the physics of the network (S1.10). Access is either set by the governor or built into the part or route (as affinity is in cells, and transport and sink kinetics are in phloem), and both kinds of setting are fixed within a mode. A setting is the rule that gives a part's access at each sensed level, so a part's access can change within a mode while its setting does not. A built-in setting is not set by the governor: it fixes the part's access within its layer, and a change in the part's requirement can move its place in the order. The governor's regulation works through stores, intake, gates and the settings it does control; where it sets a part's access, a change in requirement changes that part's draw in its own turn, not its place. When a sensed level crosses its threshold and the governor acts, as in opening a damaged part's access where it sets that access, the setting is acting, not changing: access changes and the order does not. A mode change is a change in the order, or in which levels the governor holds, other than through built-in access; it ends the snapshot. The ordered draw of Section 3.3 is the arithmetic of those settings, not a queue: a part's place in the draw is its priority, not its location.

The general model, the Persistence Allocation Model, states the architecture as follows.

In a system that regulates its own persistence, all parts draw on one shared flow of each resource, and each part's access to it is limited separately, by a setting the part does not control. A governor holds the levels the system's persistence depends on by acting through those settings; it protects the top by limiting everyone else's access first. Parts have no demand of their own. Each part works to the limit of the scarcest resource it can draw. When what a part draws falls short of what the reference state requires, whether supply falls, requirement rises or access is restricted, a shortfall is counted at that part. Where the system as a whole is short, the governor draws its stores and turns down the access of lower-ranked parts first, and they switch units off. The protected flow holds until nothing more can be taken: it shows compromise, not stress. At a fixed, fully used inflow, raising one part's access lowers another's: the resource gap is met from stores, met from outside the boundary, or left unmet at a named part, and a shortfall leaves its residue in switched-off, lost or scarred units, or in work not done. A part scales down without harm when its supply falls no faster than it can switch units off and its upkeep is still met; units are lost when supply falls faster, or below its upkeep; the part is scarred, with those units lost for good, only when the losses destroy what rebuilds it or its route back is cut. Repair is work like any other: damage raises the damaged part's requirement, which is met, like any requirement, within the part's rank.

Recovery runs the other way: after the top, the support parts' draws, the intake's among them, are served before every other part's; switched-off and lost units, and stores, come back by marginal value (S1.12). The system collapses when the protected flow can no longer be met or a non-bypassable link is cut, and dies only when no route back remains.

Two chains are real. Nothing passes between parts, but two links tie one part's state to another's:

- **dependency:** the top relies on support parts' work, so a support part's shortfall lowers delivery to the top one step later (Proposition 2): the dependency lag, counted at the top;
- **network physics:** where flow divides by physics, losing or narrowing a route raises the flow on the others (S1.10).

Cascades of work between peers are a dependency of requirement. A part whose requirement rises when another part's work falls depends on that part's work: its shortfall is counted under rising requirement, with the link documented at mapping, before outcomes, under the same guard as modes and gates (S9.2). Nothing passes between parts; the requirement link is a dependency. Most of what is called a cascade of work sits inside one part: staff are units of a team, a vacancy is a lost unit, and the team's requirement is unchanged.

3.1 Seven components

The model has seven components, and only these are called components. Everything else in this section is a mapping aid (Section 3.4).

1. **Boundary.** Fixed first. It decides everything else, including which part is the top.
2. **Governor.** A set of targets on sensed levels. If a sensed level crosses its threshold, the governor acts until it returns. It owns the signals and does no work. Its sensing and signalling are carried out by parts, which pay their cost. It is a function, not a place. The results assume its signals are true, in sensing and in command (Section 3.3).
3. **Parts.** Anything that does work. Parts have no demand of their own. Each part has a rank, fixed in two steps: its dependency layer, then its order within the layer (Section 3.3). One part is the top: the part everything else is sacrificed to keep going.
4. **Protected flow.** The flow of the resource to the top, which has to be maintained, and is being maintained, at all costs, measured against the top's need.
5. **Network.** The routes the resource moves along.
6. **Stores.** They hold resource and release it.
7. **Outside input.** Resource or energy drawn from across the boundary.

Classification rules.

- Anything that does work is a part, including work done on the governor's signal. The signal belongs to the governor and the work to the part: vessel-wall muscle narrowing a route is a part's work. Parts pay the cost of signalling.
- Repair is work like any other (Section 3.4). The intake is a part.
- Components are roles, not objects. One structure can play two roles: a vein is a route, and the blood in it is a store.
- What a part is made of are its units (cells in an organ, staff in a department), not parts.
- Where the governor sets a part's access, it sets it as a whole. It does not split a part's draw between upkeep and work: what reaches a part covers its upkeep first, then its work.

- The model describes an arrangement, not an intention: nothing in it decides, wants or chooses.

Protected flow and indicators. An indicator is whatever an observer watches. It may be the protected flow, a level the governor holds, or neither; a mapping says which and never assumes it. Proposition 2 concerns the protected flow only, and says nothing about indicators in general. Where an indicator is a level the governor holds, the governor's action holds it while its levers last.

Terminology: in the author's earlier institutional framework, "record" meant what institutions measure; that corresponds to an indicator here.

3.2 Architecture

Time runs in steps. In each step three functions apply:

$$\begin{aligned} g(t) &= G(\text{sensed state}(t), \text{environment}(t)), \\ a_{ir}(t) &= \Phi_{ir}(\{U_r, I_r, \text{stores}\}, \text{network}, g(t), \sigma_i(t)), \\ \sigma_i(t+1) &= F_i(\sigma_i(t), \{a_{ir}(t)\}_r). \end{aligned}$$

- **The governor** G holds its targets by setting access, g : store release, intake, the capacity of routes and gates, and the access of each part whose access it sets, including a damaged part's. It protects the top by limiting everyone else's access first. It does not set allocations, and it does not switch units off.
- **The network** Φ is the routes. Given the access settings and the parts' work, it gives the realised flow a_{ir} of resource r to part i . In bodies the routes are vessels and channels; in organisations, budget lines. Work that moves resource along a route is a part's work: a heart pumping, a transporter that spends energy (the receiving part's work), the finance function moving money. A drive that no part makes, such as evaporation driven by the sun, is outside input.
- **A part** F_i changes state only through what it draws, except by outside damage. It has no utility or claim. Parts have no demand of their own: what a part requires is set by a rule fixed in advance (its reference), which rises only as that rule provides: under a load applied to the system, with damage, or through a documented requirement link, never because the part asks. The governor sets access, never requirement. Damage from outside removes units or route capacity and raises the damaged part's requirement. Units lost to a shortfall are not damage.

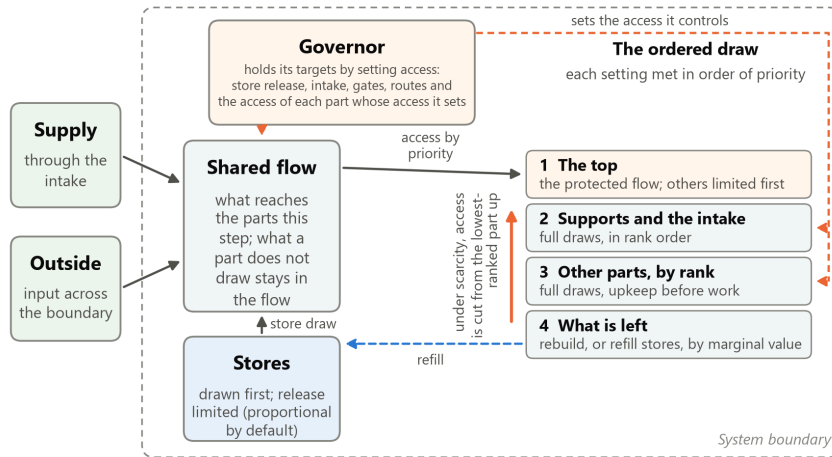
A part consists of K_i units, each **active**, **switched off** (its route back kept, basal maintenance only, little or no work), **lost** (rebuildable) or **scarred** (lost for good: the losses destroyed what rebuilds it, or its route back is cut). One part, the **top**, receives the protected flow.

3.3 The reduced form

Where the network is not modelled explicitly, the flow is written as an ordered draw (Figure 1). For resource r the available flow in a step is $S_r = U_r + I_r + \sum_s d_s$: supply through the intake, resource drawn in across the boundary, and store draws. Parts draw from it in a fixed **order** π_r , each up to its full draw:

1. the top's full need;
2. support parts' full draws, in rank order (parts the top depends on, and the intake);
3. every other part's full draw, in rank order;
4. what is left: reactivation and rebuilding against refilling the stores, by marginal value.

Each part is drawn in full in its turn, and what reaches a part covers its upkeep (basal maintenance) first, then its work and renewal. So a part that is short loses work before upkeep, and the lowest-ranked part is cut fully before the next is touched. What a part does not draw stays in the flow; nothing is returned. The ordered draw is the arithmetic of the access settings, not a queue: a part's place in the draw is its priority, not its location. A requirement raised by damage is met within the damaged part's rank, because the governor, where it sets the part's access, opens that access in response to local signals of the damage. Units lost to a shortfall are not damage, and rebuilding them is not repair. They are rebuilt from what is left: rebuilding them is not part of the part's draw, so it competes for the surplus by marginal value as supply returns (S1.12).



The resource ledger: with the flow fully used and no part above its reference: gap = store draw + outside input + shortfall at named parts. Access decides which parts go short, not how much.

Residue in units: switched off (route back kept) → lost (rebuildable) → scarred (lost for good).

Failure: collapse when the protected flow cannot be met or a non-bypassable link is cut; death only when no route back remains.

Figure 1. The model's architecture in the reduced form. The governor sets access (store release, intake, gates, route capacity and the access of each part whose access it sets), not allocations. The flow available in a step (supply through the intake, outside input and store draws) is drawn by parts in a fixed order of priority: the top's full need; support parts and the intake, in full; every other part in full, by rank, with upkeep before work inside each part; and what is left, which goes by marginal value to rebuilding or to refilling stores. Under scarcity, access is cut from the lowest-ranked part upward. The footer gives the resource ledger in the scarcity-limited regime (Proposition 1), the residue a shortfall leaves in units, and the failure conditions (S1.9 and S1.11).

Rank. Rank is fixed in two steps. The first is the dependency layer, from the top down: the top, then the parts whose work feeds the top, then the parts that feed those. The second is the order within a layer, which comes from documented access and, where access is built in, from requirement. π_r is the order of priority for r ; parts lose adequate access to r , when r alone is scarce, in the reverse order. **It is fixed in advance from documented properties of access** and, where access is built in, from the rules for requirement, and is never read off an observed order of loss. The reduced form keeps one support layer. Deeper dependency layers extend the order; the propositions are not claimed for that case.

Modes and snapshots. The top never changes within a boundary. Below it, the order follows from access and requirement. Access settings are fixed within a governor mode and change only when the mode changes. When a sensed level crosses its threshold and the governor acts, the setting is acting, not changing; access changes and the order does not. A mode change is a change in the order, or in which levels the governor holds, other than through built-in access; modes are documented independently, before outcomes are seen (S9.2). The model applies to one mode at a time, a

snapshot. A threshold switch, such as decompensation in blood loss, is a mode change: crossing its threshold changes the order, or which levels the governor holds, and ends the snapshot.

Assumptions for the results:

- results hold resource by resource;
- the ordered draw above, with a fixed order, while no requirement changes the order;
- the store release forms in Section 3.4;
- a reference fixed in advance;
- units changing only through supply, except by outside damage;
- the governor's signals are true (sensing and command): what it senses matches the real levels, and the access it sets is put into effect; the realised flow may still be limited by the network (S1.9).

3.4 Mapping aids

Everything in this subsection is a mapping aid, not a component.

Work, stores and units.

- **Work** is limited by the scarcest resource, $w_i = \min(\hat{c}_i, (1 - \epsilon_i)w_i^0, \min_r A_{ir}^w / \kappa_{ir}^w)$, with $\hat{c}_i$ effective capacity and w_i^0 reference work. The synthesising-unit form (Kooijman 2010) is the smooth general case.
- **Renewal** needs a baseline per active unit plus wear per unit of work, $\kappa^n n^a + \kappa^u w$.
- **Stores** release up to a limit in proportion to content by default, $\rho_s = k_s L_s$ per step (DEB), or at a constant rate until empty (full release). A fraction of undrawn flow refills them; the rest is spilled.
- **Units change only through supply, except by outside damage.** Unpaid basal maintenance is paid from the part's own units, so the part shrinks (DEB). Unmet renewal switches units off at up to θK per step, and the rest fail at rate $1/\tau_f$. Losses in one episode beyond a template limit $Q^* K$ are scarred, as are units whose route back is cut.
- **Units take their part's access.** Which units within a part go first (for example, by distance from the supply) is a mapping aid. Work is counted once, at the part.
- **Repair** is work like any other, of two kinds. **Routine repair**, renewing worn units, is part of each part's own draw (the renewal terms κ^n and κ^u), so it falls when that part's access falls. **Damage** raises the damaged part's requirement (rising requirement). The governor meets it as it meets any requirement, by opening that part's access where it sets that access, and by releasing stores into the flow, within the part's rank; local signals that open a damaged part's access belong to the governor. The part's response to what arrives is a consequence. Under saturable uptake, raised requirement also raises the part's adequacy threshold (S1.7), so damage can move the part's place in the order. Where the governor sets access, a raised requirement enlarges the part's draw in its own turn and does not move its rank. Repair cells working at a damaged part are counted at that part, as its units. A part whose work produces a resource that repair needs (for example, bone marrow) adds that resource to the flow, like the intake. Units lost to a shortfall are not damage, and rebuilding them is not repair: it is paid from what is left once every part has drawn (Section 3.3). No rule puts repair first or last.

The top's own routes. The top's own routes can narrow: in simulated blood loss the brain's estimated vascular resistance rose as blood velocity in its middle cerebral artery, a proxy for its blood flow, fell (Bondar et al. 1995; S7.3). The reduced form simplifies this away.

Shortfall and the two ledgers. For each part and resource, the **reference allocation** q_{ir}^0 is fixed in advance as a rule and never lowered because units have switched off. **Shortfall** is its unmet part, $\ell_{ir} = [q_{ir}^0 - a_{ir}]_+$. It has three origins: falling supply, rising requirement, or a governor mode that restricts access while resources suffice. The last is admitted only for a mode and gate documented independently in advance. Rising requirement includes requirement that rises when another part's work falls, admitted only for a requirement link documented in advance. Requirement raised by damage follows a rule fixed in advance with the reference allocation, and counts only for damage documented independently.

The resource ledger. With the gap $\Gamma_r = \sum_i q_{ir}^0 - U_r$, in the scarcity-limited regime, where the flow is fully used and no part receives above its reference, every unit of the gap is met from a store, met from outside, or left unmet at a named part (Proposition 1). In general, resource left unused and allocation above reference enter as well.

The state ledger. It records what a shortfall leaves behind: units switched off, lost or scarred, and work not done. Work not done that falls on another system, across the boundary, is outside this ledger and belongs to that system's mapping. Residue is a consequence of shortfall and is never added to the resource ledger.

The margin M is what can go unmet below the top before the protected flow moves: the other parts' full draws D where delivery to the top depends on support parts' work, and $P + D$ where it does not, with P the support parts' full draws. N is the top's full need, and $\sum_i q_i^0 = N + P + D$.

The other mapping aids are rank detail (S9.3), modes and gates (S9.2), economising, a governor mode (S1.4), templates and scars (S1.8), and viability and the capture basin (S1.11). Economising lowers access, not the reference; what goes unmet is still counted as shortfall.

3.5 Propositions

Proofs are in Appendix A. **Every quantitative statement in the main text, and every result in S1 except S1.5, S1.6 and S1.11, was checked numerically** against an independent implementation of the reduced form (Supplement S2). Two of the checks corrected earlier statements of the same results, and review corrected four more, and the final check of version 2 two more (S2, items 8 and 9); version 2 also changed the order of the draw, and its results were re-derived and rechecked.

Thirteen further results are in Supplement S1. Seven were Propositions 5 to 11 in version 1 and moved there in version 2. Their checks are unchanged, and S1.9 gains two; S1.9 and S1.13 are restated, and the proofs of S1.11 and S1.12 adjusted, for the version 2 order, and S1.8, S1.11 and S1.13 are reworded for repair as work like any other (Changes from version 1, items 9 and 11):

- rank under saturable uptake (S1.7);
- rate decides harm (S1.8);
- exhaustion, severance and constriction (S1.9);
- rerouting (S1.10);
- collapse, not death (S1.11);
- recovery order and partial refill (S1.12);
- repair under strict rank (S1.13).

The other six are a lead time (S1.1; it was G12's derivation, and is read in version 2 as the length of the G27 window), the limit under conflicting orders, shrinking, economising, the fuse, and rising requirement (S1.1 to S1.6).

Conservation

Proposition 1 (conservation under scarcity, creation by gating). In any step, $\sum_i \ell_i = \Gamma - \sum_s d_s - I + R_{\text{unused}} + X$, where R_{unused} is resource left unused (refilled, spilled or left in the flow) and X is allocation above reference.

1. **Scarcity-limited regime.** Where the flow is fully used and no part receives above its reference, total shortfall is the same under every setting of access at fixed supply, store draw and outside input. Access decides only which parts go short.
2. **Access-limited regime.** Where a gate leaves resource unused, total shortfall rises one for one with the resource left unused, balanced by refill or spill. Opening the gate by the resource it leaves unused removes that shortfall without any other part losing; opening it further is a change of access in the scarcity-limited regime.

The identity is elementary. Its content is the reference fixed in advance and never lowered, and the named exceptions: resource added, requirement lowered, a gate opened, or allocation beyond requirement taken back (Section 1). In a resource-short system, restoring one part's supply outside the named exceptions leaves an equal deficit elsewhere (Figure 2). In an access-limited system, where a gate holds a process shut while resource is available, opening the gate by the resource left unused relieves the process with no deficit elsewhere. The two regimes predict different consequences of the same intervention.

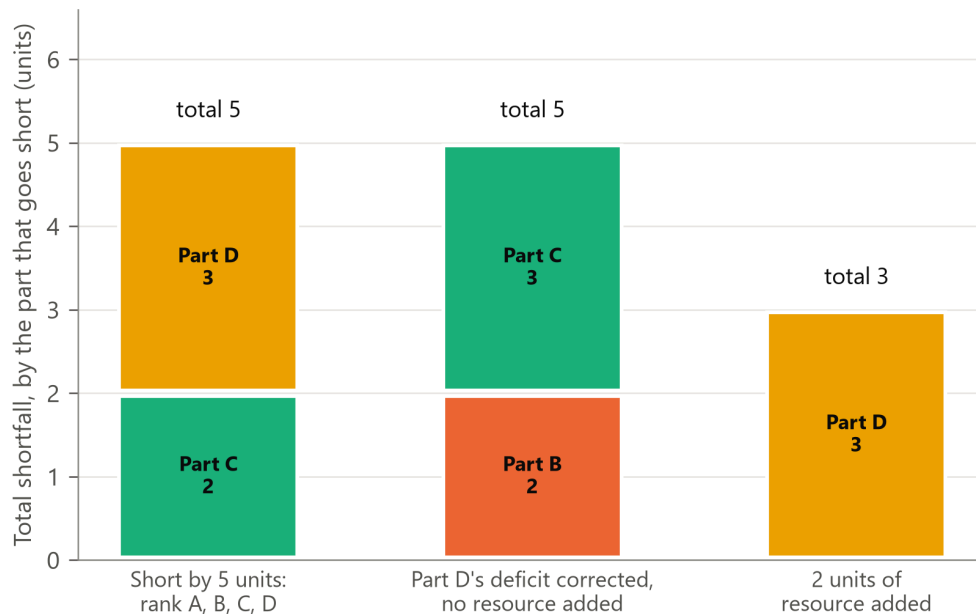


Figure 2. Proposition 1 in the scarcity-limited regime, a worked example. Four parts, A to D, each need 3 units, and supply is 7, so the system is 5 units short. Left: with rank A, B, C, D, parts C (2 units) and D (3) go short. Centre: part D's deficit is corrected by moving it to the front of the order, with no resource added; B and C go short instead, and the total stays at 5. Right: adding 2 units of resource, one of the named exceptions, lowers the total to 3.

The silence and the break

Proposition 2 (silence). In a step in which no unit of the top or a support part is still coming back from an earlier shortfall: without a delivery dependency, the protected flow is met if and only if $S \geq$

N ; where delivery to the top depends on support parts' work, the same holds if the support parts were met in the step before, and the support parts are met if and only if $S \geq N + P$. Hence, with the gap and supply the same from step to step, the protected flow holds if and only if $\Gamma \leq \sum_s d_s + I + M$, with $M = P + D$ without that dependency and $M = D$ with it.

Where the support parts were not met in the step before, the top can be short while every other part draws in full: that is the dependency lag. It is counted at the top, not a break in the order, and the earlier shortfall must be observed, not inferred from the later one (worked example in S1.9).

Where units of the top or a support part, switched off or lost in an earlier shortfall, are still coming back, the top can be short while every other part draws in full: that is the recovery lag. It is not a break in the order, and the earlier shortfall must be observed, not inferred from the later one (worked example in S2).

Inside that region the protected flow carries no information about shortfall below the top (Figure 3). Outside input extends the silence exactly as a store does. It breaks at two thresholds: at once, and one step later through dependency.

Proposition 3 (store left at the break). A constant gap Γ per step is met from one store from level L_0 , with no outside input, and $M < \Gamma \leq kL_0$. Under full release at a rate of at least Γ , the protected flow breaks when less than one step's gap is left in the store, whatever Γ (one step later with a delivery dependency). Under proportional release the margin is used up when the store falls to $L^* = (\Gamma - M)/k$: **a larger gap per step leaves more of the store unused**, linearly, with slope $1/k$. The protected flow breaks in that step without a delivery dependency, and one step later with one (Proposition 2).

The intuitive rate-independence of the break holds only for full release. Under the default release, the reserve remaining at the break should rise with the rate of loss.

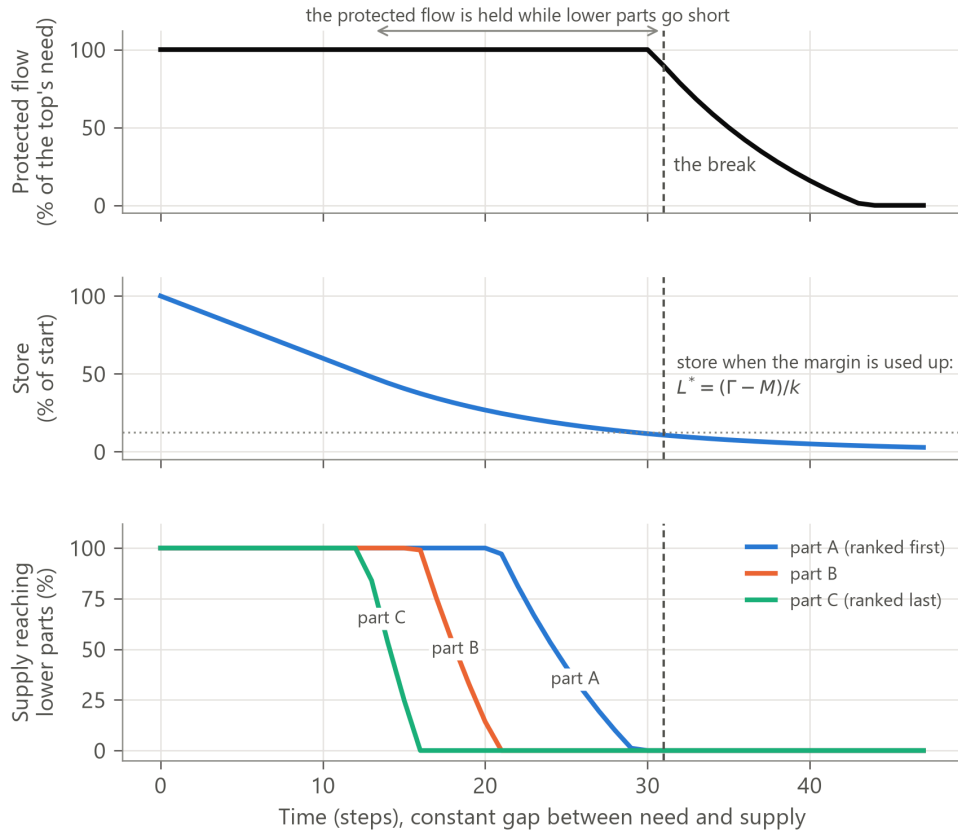


Figure 3. The silence and the break (Propositions 2 to 4), simulated in the reduced form. A constant gap of 12 units a step (need 22: the top 10, one support part 3 (upkeep 1, work 2) and three other parts at 3 each (upkeep 1, work 2); supply 10) is met from one store with proportional release ($k = 0.08$). Delivery to the top in a step is in proportion to the support part's work in the step before, so M is the other parts' full draws, 9 units. Bottom: the other parts are cut in turn, C fully before B and B fully before A (Proposition 4). Top: the protected flow stays flat while all three go short (Proposition 2). Middle: the protected flow breaks one step after the store falls to $L^* = (\Gamma - M)/k$ (through the dependency), an eighth of its starting level (Proposition 3). Parameter values are illustrative.

The order of loss

Proposition 4 (order). In a step in which the support parts were met in the step before and no unit of the top or a support part is still coming back from an earlier shortfall, parts short of their draw form a lower segment of the rank order, and a part that is short loses work before upkeep. No part goes short while a lower-ranked part draws anything. Support parts go short only after every other part has nothing, and the top goes short last.

In such a step at most one part is partly supplied. Where the support parts were not met in the step before, the top can be short while every other part draws in full: the dependency lag (Proposition 2), counted at the top, not a break in the order. Where units of the top or a support part are still coming back from an earlier shortfall, the same can happen: the recovery lag (Proposition 2).

Strict priority of this kind is an assumption in hierarchical plant models (Grossman and DeJong 1994; Marcelis and Heuvelink 2007). Here it is the reduced form of access settings, with the order fixed in advance. It is the limit case, not the general claim. What the four fields share is a ranked order of draws: under saturable uptake parts go short together when their thresholds are close (S1.7),

and in councils a coarse measure points against strict cutting (S5.2). Whether a system's priority is strict or graded is itself a test.

4. Predictions, and how they separate the model from its neighbours

The model's predictions fall into two layers. **Layer 1** is what any feedback loop that holds a flow by drawing a finite store would be expected to show: the protected flow is flat while a store is drawn (G1), the break comes at a set depletion, and a larger store gives a longer silence. Confirming layer 1 says little about the model. **Layer 2** needs selection or design: the order of loss, switches, the order of recovery, repair competition. Table 2 lists the predictions that follow from Propositions 1 to 4 alone, every prediction that has been tested, and G27, which replaces the tested G12 (its window uses S1.1). Predictions that also need a supplementary result (G5, G13 and G20 among them) are in Supplement S4, with the full list and status.

Table 2. Main predictions.

No.	Prediction	Derived from	Status
G1	The protected flow stays at the top's need while lower parts and stores move	Prop. 2	Natural observations of held indicators do not test G1. The one observation of the protected flow (by a proxy, S7.3) fell before the break and is logged against the model. H1 check (on an indicator) not consistent (low weight)
G2	Lower-ranked parts lose access first, in ascending rank	Prop. 4	Compatible: calibrated haemodynamic models (Table 4). Krieger (1921) logged against, not yet weighed (Section 6)
G3	With a store meeting the gap and its release not binding, the break comes at the same cumulative gap whatever the rate to within one step's gap (Proposition 3); with a delivery dependency, the cumulative gap at the break is larger by one step's gap, so it rises slightly with the rate. Where release binds (a tapering store, or a gap above the release rate), faster onset breaks earlier. Under proportional release, the margin is used up when the store left is (gap minus margin) divided by the store's turnover rate, so a larger gap per step leaves more of the store unused, rising linearly with the gap; the break comes then, or one step later where delivery to the top depends on the support parts	Prop. 3	Not yet tested. Indicator-defined events (a 30 mmHg fall in pressure in sheep; tree death) fit the first clause; no break of the protected flow was observed (Table 4)
G4	A larger store gives a longer silence; a depleted start breaks sooner	Prop. 2	Not yet tested. Natural observations of indicator-defined events fit (S6); they do not test the protected flow. Logged against, not yet weighed: Hikino et al. (2026; Section 6; S7.2)
G12	Recovery from small knocks slows before the break where release headroom shrinks	Supplement S1.1	Failed its first held-out test , at half weight, with two qualifications: runnable only under the pre-data reading of missing bins, and the step-down inside the fallback cohort not stated in the pre-registration (Section 5.1). Its derivation was in error; G27 replaces it

No.	Prediction	Derived from	Status
G25	The order of loss is predicted by properties of access documented beforehand	Prop. 4; S1.7	Supported in its first held-out test , at half weight (Section 5.1)
G25-C	The protection gap between lines with and without a duty widens where funding fell more (the scarcity version of G25)	Prop. 4	Not supported in its first held-out test, at half weight (Section 5.1)
G27	As the break approaches, fluctuations in the gap show first in the store's release while it has headroom, then in the access of the lowest-ranked part still supplied, then in each part above it in turn, and in the protected flow only at the break	Props. 2, 4; S1.1	Not yet tested. Added in version 2 (Section 5.1)

What the model predicts that neighbouring theories do not state. Table 3 lists results that follow in the model but are not stated in the neighbouring theories as we have read them. Some appear in words (in allostasis especially), and rerouting by physics is known in power engineering, where contingency analysis computes how flow shifts onto the surviving lines after an outage (Wood, Wollenberg and Sheblé 2014); none is stated formally across domains. Silence is not contradiction: the entries mark where the model makes a claim those theories do not make, and where a test can tell them apart.

Table 3. Results not stated formally across domains in neighbouring theories.

Result	DEB	Selfish Brain	Allostasis and allostatic load	Control theory
Under scarcity, access decides only which parts go short; gating creates and removes shortfall (Prop. 1)	Conserves mass and energy; no ledger of unmet requirement	Build-ups in front of bottlenecks; no ledger across many parts	Hidden shortfall, in words	No resource ledger
The protected flow carries no information inside the silence region (Prop. 2)	Not stated	Brain held while body loses, two compartments	Stated in words	Not stated as a shortfall budget
Store left at the break rises with the gap per step (Prop. 3)	Gives the release form; not the break	Not stated	Not stated	Not stated
Rank under saturable uptake from affinity, capacity and requirement (S1.7)	Not stated across parts	Access by insulin dependence, two compartments	Not stated	Not stated
Scar threshold in speed and depth (S1.8)	Shrinking under unpaid maintenance; no switch-off rate, unit states or scar rule; damage irreparable	Not stated	Wear, in words	Not stated
Failure of the top while parts outside it and its supports draw implies a cut or narrowed link, damage, a dependency lag or a recovery lag (S1.9)	Not stated	Not stated	Not stated	Not stated
Rerouting by physics against rerouting by settings (S1.10)	Not stated	Not stated	Not stated	Known in power-systems engineering (Wood, Wollenberg and Sheblé 2014); not stated across domains

Result	DEB	Selfish Brain	Allostasis and allostatic load	Control theory
Repair served strictly by rank (S1.13)	No repair allocation	Not stated	Repair squeezed, in words	Not stated

The sharpest discriminating tests are these:

- **Conservation:** in a resource-short system, a correction at one part that adds no resource (from outside or from a store), lowers no part's requirement, opens no gate and takes back no allocation beyond a part's requirement should leave a matching deficit elsewhere.
- **The order of loss from access** (G25) in systems the model was not built on.
- **The scar threshold:** with the route back intact, a scar needs speed and depth together (S1.8).
- **Strict-rank repair** (G23(b); S1.13).

5. Tests so far

5.1 Held-out tests

Both tests were pre-registered before any outcome data were opened. Each computation was replicated by a second AI system (ChatGPT, OpenAI): it wrote its own analysis script from the pre-registration and our code, without seeing the data or the results, and the script was run on the same files. Each verdict was then given blind against fixed adjudication rules by the same system, in a separate session with no shared context. Records are in Supplement S5.

H1: warning before falls in arterial pressure in surgical cases with heavy blood loss (G12).

- **Data:** VitalDB, an open database of high-resolution intraoperative recordings (Lee et al. 2022).
- **Prediction:** before falls in arterial pressure in cases with high blood loss, recovery from small fluctuations should slow. This was measured as a rise in the lag-1 autocorrelation of mean arterial pressure, against controls from the same case.
- **Cohorts:** the primary cohort (no vasopressor boluses) had, at 20% of estimated blood volume, 0 high-loss and 16 low-loss qualifying cases, both short of the 20 required, so it was not runnable. The fallback cohort, with a loss threshold of 15% of estimated blood volume, gave 25 high-loss and 42 low-loss cases.
- **Result:** the median excess rise was -0.072 (one-sided $p = 0.48$), and high-loss cases did not exceed low-loss ones ($p = 0.78$). **G12 failed**, at half weight, with two qualifications:
 - the test is runnable only under the pre-data reading of missing 10-second bins;
 - the step-down of the loss threshold inside the fallback cohort was fixed in code before the data were opened, but not stated in the pre-registration.
- **G1 check (low weight):** pressure stayed within 20% of baseline while haemoglobin fell in 10.7% of 75 cases, against a pre-stated "more than half". Not consistent.
- **Reading under the model's standing rule:** the first candidate for revision was the mapping, not a new mechanism. Version 2 names the faults, below. The verdict stands; G12 as stated is replaced by G27.

H1 corrected in version 2 (the verdicts stand).

- **The derivation.** G12's derivation (S1.1) put the warning in the protected flow, which Proposition 2 says is silent before the break: once the store's headroom is used up, a fluctuation is met within the margin M , below the top, and shows in the protected flow only at the break, which comes when the margin is also used up, or one step later where delivery to the top depends on the support parts, as in this mapping. H1 tested G12 as stated, and G12 as stated failed.
- **The mapping.** The corrected mapping redraws the frozen boundary (the patient's circulation plus the anaesthetist) to the patient, with the anaesthetist as an outside loop. At this whole-body boundary the top is the brain, and the protected flow is the oxygen the brain draws from its blood supply, measured against its resting need. Mean arterial pressure is an indicator. It tracks a level the governor holds: the stretch of the arterial wall at the carotid sinus and aortic arch, sensed by the baroreceptors (the arterial baroreflex; Chapleau, Hajduczuk and Abboud 1991). H1 measured an indicator, not the protected flow. This correction was made after the result: it is post hoc and is not a test (Section 6).
- **The release profile.** The frozen H1 map committed a store that tapers as it empties, possibly ending in a switch. The pre-registration named the release profile as the first candidate for revision if G12 failed, and it is logged as that. Which release profile held in the VitalDB cases is untested. The pre-registration bars explaining the null as a switch. It is not explained, and the verdict stands. A later test under anaesthesia must fix its release profile from evidence under anaesthesia: in Evans et al. (2001), anaesthetic agents blunted or abolished the compensation (for example, halothane) or the switch (for example, alfentanil).
- **G27**, defined and untested: as the break approaches, fluctuations in the gap show first in the store's release while it has headroom, then in the access of the lowest-ranked part still supplied, then in each part above it in turn, and in the protected flow only at the break. A test of G27 needs a system whose final store tapers.
- **G18 in H1.** The exploratory measure G18, pre-specified on 6 October under the original framing, paired heart rate and stroke volume with mean arterial pressure. Before falls, the correlation of pressure with heart rate became more positive (median excess change +0.126, 25 cases; window medians 0.16 to 0.33, at control times 0.20 to 0.12, from the post hoc follow-up in S5.1); with stroke volume it did not change (median excess change -0.0018, 12 cases). Heart rate moved with pressure, not against it. G18 as mapped paired two outputs of one part (the heart) with an indicator, not two parts, so it was not a test of G18 as stated, and it is not evidence for G27. It was not scored and is not counted.
- **The corrected mapping** is in a new file beside the frozen setting map, which remains the test record (S5.1).

PT1: order of loss from documented access in English local government (G25).

- **Data:** single-tier councils in England, 2014-15 to 2019-20: 121 councils and 93 spending lines.
- **Access property:** each line was classified **blind**, before any spending data were opened, by whether a statutory duty attaches to it: class A, a duty; class B, a duty of uncertain level; class C, discretionary.
- **Primary question (fixed against dynamic priority): inconclusive** at full weight. Projected client growth published before each budget showed no detectable effect on which lines were protected ($\beta = 0.05$; 95% interval -0.75 to 0.86), but the interval includes the smallest effect set in advance as meaningful (0.25).

- **G25: supported, at half weight** (the broad pattern was known in advance). Within a council and year, real spending per head on class A lines grew about 4.6 percentage points a year faster than on class C lines (one-sided $p = 2 \times 10^{-11}$; Figure 4). Class B was not distinguishable from class C, and that step fails its ordering when London is excluded. The measure is spending, the allocation each line received under its access. The frozen map set each line's reference as its base-year real spending per head; its requirement in version 2's terms, and so its shortfall in those terms, was not measured.
- **G25-C: not supported, at half weight.** The scarcity version, that the gap widens where funding fell more, was not supported: the interaction was 0.22 (SE 0.25).

This is the model's first held-out test outside bodies, and its first held-out support, for order of loss only, and for statutory over discretionary lines (A over C); the coarse, unscored measure in S5.2 points against strict-rank cutting.

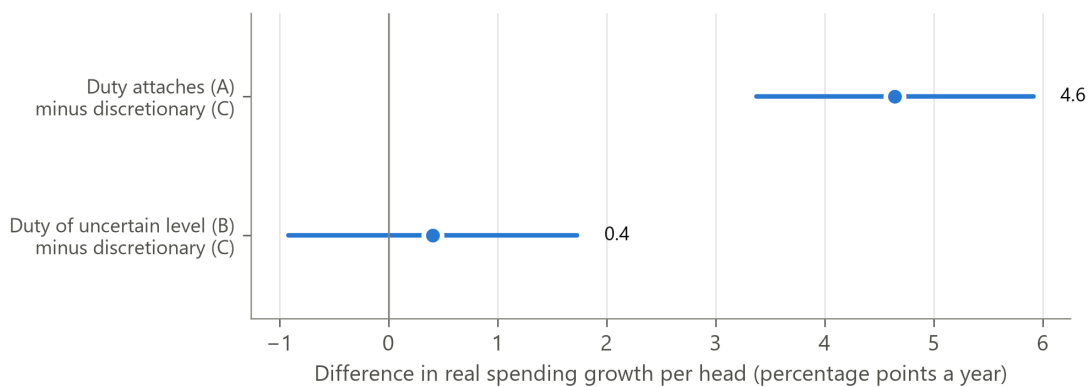


Figure 4. PT1: order of loss in English single-tier councils, 2014-15 to 2019-20 (G25; 121 councils). Difference in the growth of real spending per head between classes of spending line, within a council and year: duty (A) minus discretionary (C), 4.6 percentage points a year (one-sided $p = 2 \times 10^{-11}$); duty of uncertain level (B) minus discretionary (C), 0.4 ($p = 0.27$). Bars are approximate 95% intervals, recovered from the one-sided p-values with the analysis's t distribution (120 degrees of freedom), because the G25 results record p-values, not standard errors. The scarcity version (G25-C), not supported, is not shown.

5.2 Natural-system observations

Before the held-out tests, the model was checked against published findings in several natural systems, with predictions written down before the sources were opened. These checks are **not tests**: some outcomes were known in advance, sources were often read through summaries, and three early refinements of the model were built from blood-loss and fasting findings (Section 6). We report them as compatible observations (Table 4; full records in Supplement S6). One observation of the protected flow itself is logged against the model.

Table 4. Natural-system observations.

Feature	Blood loss	Plants in drought
Indicator steady while blood loss or drought deepens	Arterial pressure, an indicator that tracks a level the governor holds (arterial wall stretch, sensed by the baroreceptors; Section 5.1), well maintained while cardiac output falls (Evans et al. 2001); vital signs stable in early bleeding while the compensatory reserve falls (Convertino et al. 2016)	Canopy colour, an indicator not mapped here (not a flow, so not the protected flow), lags hydraulic failure, best predicting trees already dead rather than dying (loblolly pine saplings; Hammond et al. 2019)

Feature	Blood loss	Plants in drought
Protected flow held	Against the model: blood velocity in the middle cerebral artery, a proxy, fell by 27% at presyncope while arterial pressure was 2% above baseline (9 volunteers, lower-body negative pressure; Bondar et al. 1995; S7.3)	Not observed
Lower priority pays first	Splanchnic vasoconstriction the dominant compensation in a model calibrated to 35 adults under lower-body negative pressure (Bergauer et al. 2026); renal resistance rising before carotid in a model calibrated to 43 swine (Sadid et al. 2026, preprint)	Not used here: the leaves-first ordering is contested (Section 6)
Same depletion at an indicator-defined event whatever the rate	Blood volume removed at a 30 mmHg fall in mean pressure: $27.0 \pm 4.2\%$ at about 0.4% of blood volume a minute against $27.3 \pm 3.2\%$ at about 2% a minute (mean $\pm$ SE; 8 sheep, crossover; Scully et al. 2016)	Saplings died at the same loss of conductivity under fast and slow drought, with species-specific thresholds (about 95% and 45%; Dai, Wang and Wan 2018)
Switch or death at a set depletion	Decompensation (sympathetic withdrawal) once cardiac output falls to 50 to 60% of rest, about 30% blood loss (Evans et al. 2001)	About 80% loss of conductivity in loblolly pine saplings (Hammond et al. 2019)

The sheep result fits full release, or a store with turnover high relative to the gap, if the fall in arterial pressure marks the protected flow's break, which was not observed (Proposition 3). It is an observation about the release profile in that setting, not a test.

6. Scope, limits and failures

Scope.

- **Persistence is a viability constraint, not something maximised.** The governor keeps the system within the states compatible with its continued existence; within that envelope, the model says nothing about what the parts' work is for.
- **Terminal reproductive programmes are out of scope** (semelparous salmon, some marsupial males, monocarpic plants). There, individual persistence stops being the protected constraint. The model should work before the reproductive switch and fail after it.
- **The model is not a theory of action selection.**
- **The boundary, the top, the protected flow and the classification of components are fixed before outcomes** and are not redrawn after a counterexample. A mapping fault named after a result is corrected in a new file beside the frozen one, as post hoc, and is not a test. H1's corrected mapping, which redraws the frozen boundary (the patient's circulation plus the anaesthetist) to the patient, is such a correction, and the verdict stands (Section 5.1).
- **Systems with signal faults are outside the propositions.** The propositions assume the governor's signals are true (Section 3.3). For a system whose governor senses false levels, or whose settings are not carried out, the model is diagnostic only (Supplement S9.5).

Limits of the formal results.

- The propositions hold for the reduced form, not for a general network. Where several complementary resources bind together and their orders conflict, ordinal ranks do not decide the outcome (Supplement S1.2): the network must be mapped explicitly.

- **The model takes access settings as fixed within a governor mode, changing only when the mode changes, with the order below the top following from access and requirement (Section 3.3). Whether access settings change when the mode changes, and only then, is an open empirical question.** Under built-in access, a requirement that rises by its rule can move a part within its layer; the move is computed in advance (S1.7). PT1's primary question asked whether, within one setting, the order is fixed or follows projected need; its verdict was inconclusive (Section 5.1), and it is unchanged. A valid test needs two recipients competing for the same scarce resource at the same time, with the governor's mode documented, comparing the order within one mode and across a change of mode.
- The functional forms not taken from DEB (switch-off and failure rates, the scar rule) are modelling choices. No parameter has been estimated from data. Results built on these forms, the scar threshold of S1.8 among them, hold for these forms; whether they survive other plausible forms is untested.

Failures and findings against the model.

- **G12 failed** its first held-out test, at half weight and with two qualifications, and **G25-C was not supported** (Section 5.1). Version 2 finds an error in G12's derivation and in the H1 mapping; whether they caused the failure is untested, and the verdict stands.
- **Findings logged against the model, not yet weighed:**
 - the one observation of the protected flow (by a proxy): in simulated blood loss, blood velocity in the middle cerebral artery fell by 27% at presyncope, while the brain's estimated vascular resistance rose (Bondar et al. 1995; S7.3);
 - autopsies after prolonged inanition (Krieger 1921, as cited by Peters and Langemann 2009) found heart, liver, pancreas and kidney all losing about 40% of their mass while the brain lost under 2%, with no order among the organs below the brain. Whether this conflicts with the model depends on the order fixed for energy and protein at a starvation mapping;
 - in 12 Australian tree species, the predicted leaves-before-stems order of hydraulic failure (vulnerability segmentation) was "universally absent or negative" (Peters and Choat 2025);
 - in mature spruce, an earlier drought eased physiological stress in a later one, where a stressed start should fare worse (G4; Hikino et al. 2026). Leaf area fell by 60% and was still 30% lower four years later; that does not lower the canopy's requirement, since the reference is never lowered because units were lost (Section 3.4).

Details and weights are in Supplement S7.

Independence. The convergence in Section 2 rests on four fields that developed separately and does not depend on our route. Our own route is not independent of them:

- the AI system that collaborated in building the model was very likely trained on these literatures;
- some elements of the model were already known to us when they were formed;
- three early refinements were built from blood-loss and fasting findings: shared upstream supply, a threshold switch, and the point at which a store begins to release.

The dated record is in Supplement S11.

7. Discussion

Mapping a system. The model's main methodological claim is that the order of loss can be fixed before outcomes. Mapping a system (Supplement S9 in full; the corrected H1 mapping, made after H1's result, illustrates the procedure) means fixing, before any outcome data are opened, in this order:

- the boundary; then the top and the protected flow; then the viable set, and whether outside support counts;
- the resources, whether they are complementary or substitutable, their stores and release profiles;
- the governor's targets, what it acts through, and which indicators will be watched, with the kind of each (the protected flow, a level the governor holds, or neither);
- each part's units, rank for each resource (dependency layer, then documented access and, where access is built in, requirement), requirements, reference allocation, rebuild time and template limit;
- any part whose work produces a resource that repair needs (routine repair is in each part's requirements);
- the routes, and whether flow among routes divides by physics or is placed by access settings;
- any modes and gates invoked for access-limited shortfall, and any requirement links between parts, each documented independently, and the snapshot each mapping covers;
- the clock.

Anything chosen with the expected outcome in mind is declared fitted. A scenario is a load applied to a system, not a system; the top, the protected flow and the indicators are never chosen because a data set happens to record them. **Where access is by saturable uptake, the rank follows the adequacy thresholds** (S1.7): the part with the highest threshold ranks lowest and goes short first. The thresholds are computable from measured affinity, capacity and requirement.

What would count against the model:

- in a system resource-short by its own mapping, a correction at one part that adds no resource (from outside or from a store), lowers no part's requirement, opens no gate and takes back no allocation beyond a part's requirement, with no matching deficit elsewhere;
- in a held-out system, an order of loss that contradicts the order fixed in advance from dependency, documented access and, where access is built in, the rules for requirement (G25);
- harm without speed: units lost or scarred, with the route back intact and upkeep met, under a fall in renewal slower than the documented switch-off rate (S1.8);
- an ordinary system, in ordinary conditions, whose own regulation destroys recoverable viability with no higher-level system being protected.

A fifth received view. Homeostasis is usually described as balance. Under scarcity, holding the protected flow steady is paid for further down: while it holds, shortfall is counted at lower-ranked parts (Propositions 1 and 4). A level the governor holds is kept steady the same way while its levers last; we state that in words only (Section 1). Allostasis says the same of predictive regulation in words (Sterling 2012); the ledger gives it a count.

Next tests. Every prediction not yet tested can be tested; none is untestable in principle.

Prediction	What a test needs	Where the data are
G25, order of loss from access	A held-out system not known in advance to the author or the AI collaborator, with access documented before outcomes	Public budgets outside PT1's frame; hospital bed allocation; plant carbon labelling
G27, the order in which fluctuations show	A system whose final store tapers, mapped before opening, with flow or access measured part by part below the top as the break approaches	Awake volunteers under lower-body negative pressure (preferred, if the test's needs are met; see below); a haemorrhage source not opened in H1 (graded-rate animal haemorrhage series). Any retest in VitalDB counts at reduced weight
Fixed against dynamic priority	Two recipients competing for the same scarce resource at the same time, with the governor's mode documented, compared within and across a change of mode	Council spending outside PT1's frame (English single-tier councils, 2014-15 to 2019-20): a later English period, or Scottish or Welsh councils; organ perfusion studies
G3, the break and the gap per step	The break defined in advance, at two or more rates, with the store's release measured	Graded-rate haemorrhage; fast and slow drought in one species
G8, rate decides harm	Paired sudden and gradual losses of the same route	Stroke and coronary occlusion series; plant girdling
G18, co-movement before the break	Multi-organ time series, with dependencies mapped before opening	ICU databases (MIMIC-IV, eICU; credentialed)
G20, failure with receivers supplied	Network outages with routes known	Grid outage records; arterial stenosis series
G23(b), strict-rank repair	Repair measured by part after multiple injury	Trauma registries
G24, rerouting by physics against rerouting by settings	A matched pair of physically divided and budgeted networks	Grid flow redistribution; budget reallocation
G26, partial refill	Store and shortfall records over repeated episodes	Local authority reserves; seasonal fat stores in wildlife

Choosing the setting for G27. The retest of G12's corrected form, G27, prefers settings where the governor's own reflexes are intact, such as awake volunteers under lower-body negative pressure, provided the setting also meets the test's other needs: a final store that tapers, and a mapping made before opening. This is a choice of test setting, not an explanation of H1's result; H1's verdict stands.

A proposed extension treats parts with their own access loops (a tumour; a person in an institution) as nested systems; its first test, whether a tumour that only drains a host matches a host whose supply is cut by the same amount, is future work. Where signals can be wrong, including by design in institutions, the model supplies the baseline against which the fault is located (S9.5); that use is also future work.

How this work was done

The model was built in October 2026 by one author working outside academia, with an AI system (Claude, Anthropic) as collaborator for the mathematics, literature searches, numerical checks and drafting.

- **Time-stamped record:** every version of the model, every prediction and every test rule is recorded, with its commit history, in a public repository first archived on 7 October 2026 (Miller 2026; <https://doi.org/10.5281/zenodo.23222615>). Its commit history to 7 October 2026 is preserved in Software Heritage (swh:1:snp:cbdaf9b4e6c27ea5f2ce307d86ebcd49ca11fcf); later snapshots are listed in the repository's README.
- **Predictions first:** each check's and each test's predictions were committed before its sources or data were opened.
- **Tests:** each held-out test was replicated by a different AI system (ChatGPT, OpenAI), which wrote its own analysis script without seeing the data or the results, and then adjudicated blind against rules fixed in advance by the same system, in a separate session with no shared context.

The dated record of the model's development is in Supplement S11.

Changes from version 1

Version 1 was posted on SSRN on 7 October 2026 and revised there on 8 October with five figures. Its permanent copies are public record v1.2, the first posting (<https://doi.org/10.5281/zenodo.23224208>), and v1.3, version 1 as last posted (<https://doi.org/10.5281/zenodo.23241491>).

No verdict has changed:

- **H1:** G12 Fails (half weight, with two qualifications: Section 5.1); the G1 check is Not consistent (low weight).
- **PT1:** the primary question is Inconclusive; G25 is Supported (half weight); G25-C is Not supported (half weight).

Substantive changes, and why:

1. **An error in the derivation of G12, and in the H1 mapping** (Section 5.1; S5.1).
 - **The derivation:** G12 put a warning in what Proposition 2 says is silent.
 - **What H1 measured:** an indicator (arterial pressure), not the protected flow.
 - **The boundary:** the corrected mapping redraws the frozen boundary (the patient's circulation plus the anaesthetist) to the patient, with the anaesthetist as an outside loop; post hoc, not a test.
 - **The release profile:** the frozen map committed a store that tapers as it empties, possibly ending in a switch; the pre-registration named it as the first candidate for revision, and it is logged as that. Whether it affected H1 is untested.
 - **Version 1's candidate:** version 1 logged the release profile under anaesthesia, with the anaesthetist as an outside loop holding pressure, as the candidate for revision, and required any explanation to name a fault and be tested in advance. The release profile stays logged as that candidate, untested; the anaesthetist is now outside the boundary in the corrected mapping, post hoc. G12 as stated is replaced by G27 because its derivation was in error (S1.1, note); its failure stands.
 - **What follows:** the corrected prediction, G27, is added untested; G18's result in H1 is reported in full; the corrected mapping is in a new file.
 - **Why:** the error was found after version 1. A correction is published as a new version, with nothing redacted.
2. **Seven components** (Section 3.1). Only these are components; everything else is a mapping aid. The governor is now a set of targets on sensed levels that owns the signals and does no work; the network is routes only, and work that moves resource is a part's (a transporter that spends energy is the receiving part's work).
 - **Why:** to separate what the model is from the aids used to map it, so that a mapping cannot add components to fit a case.
3. **"Record" retired; the protected flow and indicators** (Section 3.1).
 - **What changed:** in version 1, "record" meant the ratio of the top's output to its level, and the H1 mapping used it for the figure observers watch. The protected flow is now the flow of the resource to the top, against the top's need; an indicator is whatever an observer watches.
 - **Why:** "record" ran the two together, which is how H1 came to test an indicator. A guard is added: the top, the protected flow and the indicators are never chosen because a data set records them (Section 7; S9).

4. **The architecture stated in plain words** (Section 3): one shared flow; access limited per part by settings the parts do not control; the top protected by limiting everyone else's access first; the ordered draw as the arithmetic of the settings, not a queue. Rank is fixed in two steps, the dependency layer and then the order within a layer, with modes and snapshots (Section 3.3). π_r is now the order of priority, with loss in the reverse order (S9.3 corrected to match), and the propositions are not claimed beyond one support layer. The state ledger no longer counts work not done that another system must then do; that work belongs to the other system's mapping. The access function Φ now takes outside input I_r and the parts' work. Access is set by the governor or built into the part or route, and both kinds of setting are fixed within a mode (a setting is the rule that gives a part's access at each sensed level, so access can change within a mode while the setting does not); under built-in access a change in requirement can move a part within its layer, and where the governor sets access it changes only that part's draw. A setting acting changes access, not the order; a mode change ends the snapshot.
 - **Why:** to make the order a matter of priority, with access settings fixed within a mode, and to say when the order may change.
5. **The main text holds Propositions 1 to 4.** Propositions 5 to 11 are now S1.7 to S1.13, with their checks unchanged (S1.9 and S1.13 restated, and the proofs of S1.11 and S1.12 adjusted, for the version 2 order, item 9; S1.8, S1.11 and S1.13 reworded for repair, item 11), and their figure is now Figure S1. Table 2 now lists only the predictions that follow from Propositions 1 to 4, the tested predictions and G27; the rest are in Supplement S4.
 - **Why:** to keep the main text to the law and its direct consequences.
6. **Table 4 and G1.** Table 4's first row now reports indicators, and a new row logs the one observation of the protected flow (by a proxy), which fell before the break (Bondar et al. 1995). G1's status says so. The statuses of G2, G3, G4 and G6, Table 4's rows 3 and 4, S6's summary and S7.3 are re-read in the same way.
 - **Why:** under the new terms, the earlier observations were of indicators.
7. **Notation:** symbol clashes removed (S1.0).
8. **Shortfall, not load; no movement language.** The unmet part of the reference is now called the shortfall, and the law is stated without movement: at a fixed, fully used inflow, raising one part's access lowers another's, and, while no part receives more than it requires, access decides only which parts go short. The law names four exceptions; the fourth, taking back what a part receives beyond what it requires, matches Proposition 1. Cascades of work between peers are counted as rising requirement through a dependency named in advance.
 - **Why:** a shortfall is a count at a part, not something that moves; version 1's "load is relocated" suggested otherwise.

9. **Phase order.** Version 1 drew every part's basal maintenance before the support parts' work.

Version 2 draws the top, then the support parts in full, then every other part in full, in rank order; what reaches a part covers its upkeep first. Propositions 2 to 4, S1.1 and G27 were re-derived and rechecked (S2). G5, G13, G16, G20, S1.9 and S1.13 are restated, and G5 and G20 are now placed in layers 1 and 2. The proofs of S1.11 and S1.12 are adjusted, and S1.12's statement, S1.5, G6 and G22 now say that the support parts' draws, the intake's among them, are served before every part outside the supports, and that their switched-off and lost units come back by marginal value (version 1: the intake comes back first, whatever its rank). S1.3 gains a sentence (upkeep goes short only once the work draw is cut to nothing), and S1.6 a condition (what reaches the part at its turn). Proposition 1 is rechecked in the version 2 order, and economising is now a governor mode. The simulation results come from the development engines, which ran version 1's order or earlier orders, each with its own recovery rule, and have not been rerun under version 2's order. S4 now labels them "simulation result in a development engine (version 1's order or earlier)"; withdraws the label from G2, G5, G13 and G20, whose results depend on the order; and withdraws it from G6, G10 and G22, which an engine built into its rules, so they are assumptions, not results. Version 1 labelled them as simulation results without these limits.

- **Why:** rank runs from the top down by dependency (CANON 4), so each part is drawn in full in its turn, and upkeep comes first only within a part; where the governor sets a part's access, it sets it as a whole.

10. **Signal integrity** (Sections 3.1, 3.3, 6 and 7; S9.5). The results assume the governor's signals are true, in sensing and in command. A system with a signal fault is outside the propositions, and for it the model is diagnostic only: reverse mode lists three signatures of a signal fault, under a guard. The next test of G27 prefers settings where the governor's own reflexes are intact.

- **Why:** version 1 did not state the assumption. Stating it marks where the propositions stop; the guard keeps a signal fault from rescuing a failed prediction, and H1's verdict stands.

11. **Repair is work like any other** (Section 3.4; S1.8, S1.11 and S1.13; G23). Version 1 gave repair to a repair network and claimed repair is cut first (Section 2; Table 1). Version 2 states no such rule: routine repair is part of each part's own draw, and damage raises the damaged part's requirement, which is met within the part's rank. A part whose work produces a resource that repair needs is a part like the intake, and repair cells working at a damaged part are its units. S1.13 is restated as the ordered draw with a limited repair resource as the flow, with its proof unchanged; G23 is restated to match; Table 1's column is now "Repair competes for access". Where the governor sets a damaged part's access, opening it is the governor's fixed setting acting, so damage does not end a snapshot. Units lost to a shortfall are not damage, and rebuilding them is not repair.

- **Why:** repair draws on the same flow, under the same access, as any other work, so no rule puts it first or last.

12. **Arrangement, not intention; no part death** (Sections 1, 3, 6 and 7; Table 3; Supplement S1, S2, S4, S9 and S11). The model describes an arrangement. Wording that gave the governor, parts or system an intention ("has learned", "pursues its purposes", "willing receivers", "by choice", "remembered") is restated as an arrangement or a consequence. The central claim no longer says a part dies: units are lost for good when the losses destroy what rebuilds them or the route back is cut.

- **Why:** nothing in the model decides, wants or chooses, and the system dies, not its parts.

13. **Neighbouring theories added** (Section 2; S3): Ashby's essential variables (Ashby 1952), the free energy principle (Friston 2010, 2013), and its link to the Selfish Brain (Peters, McEwen and Friston 2017). S3 now compares twenty theories. S3 and Table 1 state a bold rule (no other theory formal in that column), and the claims about them were corrected; allostatic load and DEB are blank for systems other than bodies.
 - **Why:** an early form of the governor, and a formal account of persistence, were missing from the comparison.
14. **The dependency lag and the recovery lag** (Propositions 2 and 4; S1.9; G20; S9.5). The propositions are stated for steps in which the support parts were met in the step before and no unit of the top or a support part is still coming back from an earlier shortfall. Where the support parts were not met, the top can be short while every other part draws in full: the dependency lag, counted at the top, not a break in the order. Where units of the top or a support part are still coming back, the same can happen: the recovery lag, which is not a break in the order either. In both, the earlier shortfall must be observed, not inferred. Worked examples are in S1.9 (the dependency lag) and S2 (the recovery lag), with numerical checks in S2.
 - **Why:** the final check found that each lag broke the earlier statements.
15. **Corrections of version 1's statements:** the count of review corrections (two became four, and the final check of version 2 added two: S2, items 8 and 9); G1 now includes outside input; G3 says the store left rises linearly with the gap; S1.7 gains a sentence on built-in access, and its statement on strict priority is corrected (priority approaches strict; S9.3 likewise); S1.10's example sentence and G24's status line were removed as unsourced; S11.4 says outside reviews are logged in the author's working records, and that two imports are not cited (the newsvendor critical fractile and Michaelis-Menten competition); S11.5 says the author's route as a whole is not independent of the four fields. DEB: three of its forms are imported, not four (Section 2); Table 3's DEB entry for scars is corrected; "harm needs speed and depth" became "a scar needs speed and depth" (Section 4). The test records: H1's primary cohort is named by its frozen criterion, no vasopressor boluses, not "no fluid boluses" (Section 5.1; S5.1); S5.1 now discloses a procedural deviation (the replication ran before the adjudication); S5.2 describes D-1 correctly (21 data names that differ from the 2025-26 guidance names). Section 1 and the abstract: novelty is claimed against the theories compared in Table 3; "a property of the systems" became "not a convenience of one model"; the opening examples are restated with sources, and resilience is no longer said to be borrowed from the system's own parts. Hikino et al. (2026): version 1 counted the finding as explained, by a scar that reduces demand; version 2 withdraws that explanation, since the reference is never lowered because units were lost, and logs the finding against G4, not yet weighed (Section 6; S7.2). The power-engineering precedent for rerouting is now cited (Section 4).
 - **Why:** each was wrong, unsourced or out of date, and a correction is stated, not made silently.

Appendix A. Proofs of the main-text propositions

Proposition 1. Conservation within the step gives $U + I + \sum_s d_s = \sum_i a_i + R_{\text{unused}}$. With $\ell_i = [q_i^0 - a_i]_+$, $\sum_i (q_i^0 - a_i) = \sum_i \ell_i - X$. Hence $\sum_i \ell_i = \sum_i q_i^0 - U - I - \sum_s d_s + R_{\text{unused}} + X$, the identity stated. Regime 1: $R_{\text{unused}} = 0$ and $X = 0$, so the total contains no access term. Regime 2: $X = 0$ and R_{unused} equals the resource the gate leaves unused. ■

Proposition 2. The top draws first, then the support parts' full draws, then every other part's full draw, each in rank order. While no unit of the top or a support part is still coming back from an earlier shortfall, their draws are their full needs, so the top's draw is met if and only if $S \geq N$, and the supports' if and only if $S \geq N + P$. Delivery to the top can then fall short of its draw only through the dependency on supports, which is excluded when they were met in the step before; with the gap and supply the same from step to step, they are met in every step if they are met in one. Where units of the top or a support part are still coming back, their draws are below their needs, and the top can be short with $S \geq N$: the recovery lag. Since $\sum_i q_i^0 = N + P + D$ and $S = U + I + \sum_s d_s$, $S \geq N + P$ is equivalent to $\Gamma \leq \sum_s d_s + I + D$, and $S \geq N$ to $\Gamma \leq \sum_s d_s + I + P + D$. ■

Proposition 3. While $kL \geq \Gamma$ the store releases the whole gap, so L falls by Γ per step. Once $kL < \Gamma$ the store releases kL , L falls by the factor $(1 - k)$ per step, and $\Gamma - kL$ goes unmet below the top. The margin is used up at the first step with $\Gamma - kL > M$, that is $L < (\Gamma - M)/k$; in discrete time the level then lies in $[(1 - k)L^*, L^*)$. By Proposition 2 the protected flow breaks in that step without a delivery dependency, and one step later with one, the store then lying in $[(1 - k)^2 L^*, (1 - k)L^*)$. Under full release with rate at least Γ , the store meets the whole gap until empty, independent of Γ . ■

Proposition 4. In a step in which the support parts were met in the step before and no unit of the top or a support part is still coming back from an earlier shortfall, delivery to the top is its draw, and the draws of the top and the supports are their full needs. Parts draw in turn, highest rank first, each in full. A part is short only if the flow is used up at its turn, so every lower-ranked part draws nothing. Within a part, what reaches it covers its upkeep before its work. The top draws first, and the supports next. ■

References

- Ames BN (2006). Low micronutrient intake may accelerate the degenerative diseases of aging through allocation of scarce micronutrients by triage. *PNAS* 103:17589-17594. <https://doi.org/10.1073/pnas.0608757103>
- Ashby WR (1952). *Design for a Brain*. Chapman & Hall, London (read in the 1954 reprint with corrections, Wiley, New York).
- Bergauer A, Urevc J, Halilović M, Batzel J, Pivec V, Goswami N (2026). Modeling sex-dependent cardiovascular responses to lower body negative pressure. *Am J Physiol Heart Circ Physiol* 331:H240-H259. <https://doi.org/10.1152/ajpheart.00258.2026>
- Bobba-Alves N, Juster R-P, Picard M (2022). The energetic cost of allostasis and allostatic load. *Psychoneuroendocrinology* 146:105951. <https://doi.org/10.1016/j.psyneuen.2022.105951>
- Bondar RL, Kassam MS, Stein F, Dunphy PT, Fortney S, Riedesel ML (1995). Simultaneous cerebrovascular and cardiovascular responses during presyncope. *Stroke* 26:1794-1800. <https://doi.org/10.1161/01.str.26.10.1794>
- Buttgerit F, Brand MD (1995). A hierarchy of ATP-consuming processes in mammalian cells. *Biochem J* 312:163-167. <https://doi.org/10.1042/bj3120163>
- Chappleau MW, Hajduczuk G, Abboud FM (1991). Paracrine role of prostanoids in activation of arterial baroreceptors: an overview. *Clin Exp Hypertens A* 13:817-824. <https://doi.org/10.3109/10641969109042085>
- Convertino VA, Hinojosa-Laborde C, Muniz GW, Carter R III (2016). Integrated compensatory responses in a human model of hemorrhage. *J Vis Exp* 117:54737. <https://doi.org/10.3791/54737>

- Dai Y, Wang L, Wan X (2018). Relative contributions of hydraulic dysfunction and carbohydrate depletion during tree mortality caused by drought. *AoB Plants* 10:plx069. <https://doi.org/10.1093/aobpla/plx069>
- Evans RG, Ventura S, Dampney RA, Ludbrook J (2001). Neural mechanisms in the cardiovascular responses to acute central hypovolaemia. *Clin Exp Pharmacol Physiol* 28:479-487. <https://doi.org/10.1046/j.1440-1681.2001.03473.x>
- Friston K (2010). The free-energy principle: a unified brain theory? *Nat Rev Neurosci* 11:127-138. <https://doi.org/10.1038/nrn2787>
- Friston K (2013). Life as we know it. *J R Soc Interface* 10:20130475. <https://doi.org/10.1098/rsif.2013.0475>
- Göbel B, Langemann D (2011). Systemic investigation of a brain-centered model of the human energy metabolism. *Theory Biosci* 130:5-18. <https://doi.org/10.1007/s12064-010-0105-9>
- Grossman YL, DeJong TM (1994). PEACH: a simulation model of reproductive and vegetative growth in peach trees. *Tree Physiol* 14:329-345. <https://doi.org/10.1093/treephys/14.4.329>
- Hammond WM, Yu K, Wilson LA, Will RE, Anderegg WRL, Adams HD (2019). Dead or dying? Quantifying the point of no return from hydraulic failure in drought-induced tree mortality. *New Phytol* 223:1834-1843. <https://doi.org/10.1111/nph.15922>
- Hikino K, Hesse BD, Gebhardt T, Hafner BD, Buchhart C, et al. (2026). Drought legacy in mature spruce alleviates physiological stress during recurrent drought. *Plant Biol* 28:637-648. <https://doi.org/10.1111/plb.70039>
- Hochachka PW, Buck LT, Doll CJ, Land SC (1996). Unifying theory of hypoxia tolerance: molecular/metabolic defense and rescue mechanisms for surviving oxygen lack. *PNAS* 93:9493-9498. <https://doi.org/10.1073/pnas.93.18.9493>
- Kleinrock L (1965). A conservation law for a wide class of queueing disciplines. *Naval Res Logist Q* 12:181-192. <https://doi.org/10.1002/nav.3800120206>
- Kooijman SALM (2010). *Dynamic Energy Budget Theory for Metabolic Organisation*, 3rd edn. Cambridge University Press. <https://doi.org/10.1017/CBO9780511805400>
- Krieger M (1921). As cited in Peters and Langemann (2009); original not read.
- Lee H-C, Park Y, Yoon SB, Yang SM, Park D, Jung C-W (2022). VitalDB, a high-fidelity multi-parameter vital signs database in surgical patients. *Sci Data* 9:279. <https://doi.org/10.1038/s41597-022-01411-5>
- Marcelis LFM, Heuvelink E (2007). Concepts of modelling carbon allocation among plant organs. In: Vos J, Marcelis LFM, de Visser PHB, Struik PC, Evers JB (eds) *Functional-Structural Plant Modelling in Crop Production*. Springer, pp 103-111. https://doi.org/10.1007/1-4020-6034-3_9
- Miller J (2026). Record for "From cells to councils: a conservation law of allocation under scarcity" (dataset). Zenodo. <https://doi.org/10.5281/zenodo.23222615> (all versions; v1.0 is <https://doi.org/10.5281/zenodo.23222616>)
- Minchin PEH, Thorpe MR, Farrar JF (1993). A simple mechanistic model of phloem transport which explains sink priority. *J Exp Bot* 44:947-955. <https://doi.org/10.1093/jxb/44.5.947>
- Noakes TD (2012). Fatigue is a brain-derived emotion that regulates the exercise behavior to ensure the protection of whole body homeostasis. *Front Physiol* 3:82. <https://doi.org/10.3389/fphys.2012.00082>

- Peters A, Schweiger U, Pellerin L, Hubold C, Oltmanns KM, Conrad M, Schultes B, Born J, Fehm HL (2004). The selfish brain: competition for energy resources. *Neurosci Biobehav Rev* 28:143-180. <https://doi.org/10.1016/j.neubiorev.2004.03.002>
- Peters A, Langemann D (2009). Build-ups in the supply chain of the brain: on the neuroenergetic cause of obesity and type 2 diabetes mellitus. *Front Neuroenergetics* 1:2. <https://doi.org/10.3389/neuro.14.002.2009>
- Peters A, McEwen BS, Friston K (2017). Uncertainty and stress: why it causes diseases and how it is mastered by the brain. *Prog Neurobiol* 156:164-188. <https://doi.org/10.1016/j.pneurobio.2017.05.004>
- Peters JMR, Choat B (2025). Out on a limb: testing the hydraulic vulnerability segmentation hypothesis in trees across multiple ecosystems. *Plant Cell Environ* 48:2162-2177. <https://doi.org/10.1111/pce.15249>
- Powers WT (1973). Feedback: beyond behaviorism. *Science* 179:351-356. <https://doi.org/10.1126/science.179.4071.351>
- Sadid S, Eden MJ, Mobin FU, Gomez MK, Januszko S, et al. (2026). Calibration of a closed-loop model of porcine aortic hemodynamics during hemorrhage. *bioRxiv* preprint. <https://doi.org/10.64898/2026.01.30.702699>
- Scully CG, Daluwatte C, Marques NR, Khan M, Salter M, Wolf J, et al. (2016). Effect of hemorrhage rate on early hemodynamic responses in conscious sheep. *Physiol Rep* 4:e12739. <https://doi.org/10.14814/phy2.12739>
- Shephard RJ (2009). Is it time to retire the "central governor"? *Sports Med* 39:709-721. <https://doi.org/10.2165/11315130-000000000-00000>
- Sprengell M, Kubera B, Peters A (2021a). Brain more resistant to energy restriction than body: a systematic review. *Front Neurosci* 15:639617. <https://doi.org/10.3389/fnins.2021.639617>
- Sterling P (2012). Allostasis: a model of predictive regulation. *Physiol Behav* 106:5-15. <https://doi.org/10.1016/j.physbeh.2011.06.004>
- Straub RH (2014). Insulin resistance, selfish brain, and selfish immune system: an evolutionarily positively selected program used in chronic inflammatory diseases. *Arthritis Res Ther* 16(Suppl 2):S4. <https://doi.org/10.1186/ar4688>
- Wood AJ, Wollenberg BF, Sheblé GB (2014). *Power Generation, Operation, and Control*, 3rd edn. Wiley, Hoboken, NJ (contents read: Chapter 7, contingency analysis; Appendix 7B, line outage distribution factors).

Supplementary material

S1. Notation, supplementary propositions and proofs

What this section holds:

- **S1.0:** the notation for the whole paper.
- **S1.1 to S1.6:** as in version 1, apart from the notation (S1.0), the version 2 note under S1.1, one added sentence in S1.3, S1.4 restated with economising as a governor mode, S1.5 restated for the intake as a support part, and a condition added to S1.6 (what reaches the part at its turn).
- **S1.7 to S1.13:** the former main-text Propositions 5 to 11, moved here in version 2. Their numerical checks (S2) are unchanged, and S1.9 gains two. Apart from the notation (S1.0) and the version 2 terms: S1.7 gains a sentence on built-in access and its statement on strict priority is corrected; S1.10 drops an unsourced example; S1.9 and S1.13 are restated, the proofs of S1.11 and S1.12 adjusted, and S1.12's statement restated for the intake as a support part, for the version 2 order; S1.9 is also restated for the dependency lag, with a worked example; S1.8, S1.11 and S1.13 are reworded for repair as work like any other.

Main-text Propositions 1 to 4 are cited by their numbers.

S1.0 Notation

Version 2 removes the symbol clashes of version 1; each change is listed in the last column.

Symbol	Meaning	Change from version 1
$t; i, j; r; s$	Step; parts; resource; store	
$G; g$	The governor; the access settings it sets	
Φ	The network's flow function: given access and the parts' work, the realised flow along the routes (the network)	
$F_i; \sigma_i$	Part i 's state function; part i 's state	σ_i was s_i , which clashed with the store index
π_r	Order (priority) for resource r	
$U_r; I_r; d_s$	Supply through the intake; outside input; draw on store s	
$S_r = U_r + I_r + \sum_s d_s$	Flow available in a step	
$L_s, L; L_0; k_s, k; \rho_s = k_s L_s$	Store level; starting level; release constant; release limit of store s under proportional release (it releases up to ρ_s per step)	
$q_{ir}^0; a_{ir}; \ell_{ir} = [q_{ir}^0 - a_{ir}]_+$	Reference allocation; allocation; shortfall	
$\Gamma_r = \sum_i q_{ir}^0 - U_r$	The gap	
$R_{\text{unused}}; X$	Resource left unused; allocation above reference	
$N; P; D$	The top's full need; the support parts' full draws (upkeep and work); the other parts' full draws	D replaces version 1's B (all basal maintenance) and D_4 (ordinary parts' work), with the version 2 order
M	The margin: what can go unmet below the top before the protected flow moves	
$L^* = (\Gamma - M)/k$	Store when the margin is used up (Proposition 3)	Kept; the refill level now has its own symbol

Symbol	Meaning	Change from version 1
$L_w = \Gamma/k$; T_{lead}	Store level at which release headroom reaches zero; lead time (S1.1)	
K_i	Units of part i , in all states	Kept for units
n_i ; n_i^a	Units present (active or switched off); active units	Defined; version 1 used n in S1.3 without definition
θ ; θ_{re}	Switch-off share per step (θK units a step); reactivation share	
τ_f ; Q^*	Failure time; template limit as a share ($Q^* K$ units)	
φ ; Δ ; Λ ; x	Speed and depth of a fall in renewable units; units lost; excess of active over renewable units (proof of S1.8)	
w_i ; w_i^0 ; $\hat{c}_i$; ϵ	Work; reference work; effective capacity; economising share	
A_{ir}^w ; κ_{ir}^w ; κ^n ; κ^u ; κ^b	Resource reaching work; requirement per unit of work; baseline renewal per active unit; wear per unit of work; basal maintenance per unit	
a^b ; u ; m ; y ; χ	Basal allocation; units broken down; resource recoverable per unit; overhead; funded share of basal maintenance (S1.3)	χ was β , which clashed with PT1's coefficient
C ; v_i ; V_i ; K_i^M	Pool level; uptake; maximum uptake; half-saturation constant (S1.7)	K_i^M was K_i , which clashed with units
$\eta_i = q_i^0/V_i$; C_i^*	Reference requirement as a share of maximum uptake; adequacy threshold (S1.7)	η_i was r_i , which clashed with the resource index
Ψ ; c_S ; ν ; ω ; L^*	Distribution of past episode depths; cost per unit of a store being short; expected frequency of shortfall; best competing value of a part's binding units; refill level (S1.12)	Were F , c_S , ρ , V , L^* : four clashes removed
W ; δ_j	The repair resource per step; the total requirement raised by damage to part j (S1.13)	δ_j was D_j , which clashed with version 1's D_4
e ; z	A route (S1.10); a split (S1.2)	

Kept, and distinct: M (the margin) and m (S1.3); w_i (work) and W (the repair resource); $\hat{c}_i$ (capacity) and c_S (a cost); G and g ; C and C_i^* ; X (allocation above reference) and x (S1.8's excess); N and n_i ; U and u (S1.3); I and the index i ; S and the index s ; Φ and φ ; ν and v_i ; ω and w_i . Each pair is a different letter or case, defined where it is used.

S1.1 The G27 window (formerly: warning before the break), and its length. Under proportional release ($\rho = kL$), with no outside input, and a constant gap Γ with $M < \Gamma \leq kL_0$:

- release headroom $kL - \Gamma$ falls linearly in time while the store meets the gap, and reaches zero at $L = \Gamma/k$. From then on, recovery from small perturbations of the protected flow cannot draw on the store and slows [withdrawn in version 2: see the note];
- the margin is used up after a lead time $T_{\text{lead}} \approx \ln(\Gamma/(\Gamma - M))/(-\ln(1 - k)) \approx k^{-1} \ln(\Gamma/(\Gamma - M))$, which shortens as the gap grows relative to the margin and vanishes as the margin goes to zero. The break follows in that step without a delivery dependency, and one step later with one;
- under full release, headroom is constant until the store holds less than one step's release, so there is no slowing before the store empties.

Proof. Headroom is zero at $L_w = \Gamma/k$. Thereafter L decays by the factor $(1 - k)$ per step until $L < (\Gamma - M)/k$ (Proposition 3); the number of steps is $\ln(L_w/L^*)/(-\ln(1 - k))$. ■

Note (version 2). The first bullet put the slowing in the protected flow, which Proposition 2 says is silent before the break: once the store's headroom is used up, a fluctuation is met within the margin M , below the top, not from the protected flow. G12 was derived from this bullet and failed its first

held-out test (main text, Section 5.1); the derivation is kept here as the statement that was tested. The arithmetic stands. Read in version 2's terms, the lead time is the length of the G27 window: the number of steps, between the store's headroom running out and the break, during which fluctuations in the gap show in the access of parts below the top (G27). With a delivery dependency the window is one step longer than the lead time. This reading is new in version 2.

S1.2 The limit of rank under joint scarcity. Two parts each need one unit of each of two complementary resources per unit of work; one unit of each is available; the orders conflict (part A first for resource 1, part B first for resource 2).

- Allocating each resource by its own order gives neither part any work.
- Every split $(z, 1 - z)$ of both resources gives total work 1 and is Pareto efficient. The two ordinal orders do not settle which of them results.

Proof. Under the law of the minimum, allocation by order gives A both units of resource 1 and none of resource 2, and B the reverse. Any common split gives work z and $1 - z$. ■ Consistent with multi-resource allocation under Leontief preferences (Ghodsi A, Zaharia M, Hindman B, Konwinski A, Shenker S, Stoica I (2011), Dominant resource fairness: fair allocation of multiple resource types, Proceedings of the 8th USENIX Symposium on Networked Systems Design and Implementation). Where this case arises, the order of loss is set by the network, which must be mapped.

S1.3 Shrinking when upkeep goes unpaid. A part's upkeep is short only once its work draw has been cut to nothing, since what reaches a part covers its upkeep first. A part whose basal maintenance is short pays from its own units and loses $u = [\kappa^b n - a^b]_+ / (\kappa^b + m/y)$ of them, where m is the resource recoverable per unit broken down and $y \geq 1$ the overhead. The remaining units are exactly funded, and $u \leq n(1 - \chi)$ (the unfunded share, with χ the funded share), with equality when units hold nothing usable.

Proof. u solves $\kappa^b(n - u) = a^b + (m/y)u$. ■ This is the DEB shrinking rule with absolute preference for reserve (Kooijman 2010, Sections 4.1.5 and 3.7.4, eq. 4.6).

S1.4 Economising loses no units. In the economising mode (a governor mode, documented in advance; not the default order), the access of every part outside the top and its supports is lowered by a share ϵ of its work draw, so work is cut evenly while what reaches each part still covers its upkeep first; it does not cut renewal. It therefore causes no unit loss at any speed. Its saving arrives at once for the work and its wear, $\sum_i (\kappa_i^w + \kappa_i^u) w_i^0 \epsilon$ per step, and as idle units are switched off for their baseline renewal. Economising lowers access, not the reference; what goes unmet is still counted as shortfall. The reduced form serves the top's full need first and keeps it out of economising. Whether a top cuts its own work to keep its supports going, as neurons of the anoxia-tolerant turtle suppress their own firing (Hochachka et al. 1996), is left to a later version.

Proof. Units are lost only through unmet basal maintenance or renewal. Economising lowers the work draw, and with it the wear part of renewal need; the baseline need of units still active stays funded until they are switched off. ■

S1.5 The fuse. The lowest-ranked part is the first to go short (Proposition 4). It is the last to come back only if its marginal value in recovery is also lowest, since recovery allocates by marginal value (S1.12). The intake's draw is the exception: it is a support part, so after refeeding its draw is served before every part outside the supports; its switched-off and lost units come back by marginal value, like any other part's.

S1.6 Rising requirement. For a part with switched-off units, rising requirement is met first within active capacity, then by reactivation at up to $\theta_{\text{re}}K$ per step at a cost. Output falls short only when the rise outpaces reactivation, reactivation cannot be paid for, requirement exceeds total capacity, or what reaches the part at its turn falls short of the raised requirement (Proposition 4).

Proof. Work is bounded by active capacity and by what reaches the part (main text, Section 3.4); reactivation is the only route from switched off to active, and it is rate-limited and paid from what is left in the flow. ■

S1.7 Rank under saturable uptake (Proposition 5 in version 1). Parts take a shared resource by saturable uptake, $v_i = V_i C / (K_i^M + C)$, from a pool at level C . Let $\eta_i = q_i^0 / V_i$ be each part's reference requirement as a share of its maximum uptake. Part i is adequately supplied if and only if $C \geq C_i^* = K_i^M \eta_i / (1 - \eta_i)$, so parts lose adequate access in descending order of C_i^* . Affinity (K_i^M) alone decides the order only where η_i is equal across parts. Priority approaches strict where neighbouring thresholds are far apart (S2), and parts go short together where they are close.

Rank is then a measurable property, fixed before outcomes. It combines affinity, capacity and requirement: a high-affinity part working near its maximum can lose adequate supply before a low-affinity part with large capacity and a small requirement. This connects the model's rank to Michaelis-Menten competition, to measured hierarchies of ATP consumers (Buttgereit and Brand 1995) and to triage by binding affinity (Ames 2006). It is the formal case of access built into the parts: a built-in setting is fixed within a mode, and a change in a part's requirement can move its place (main text, Section 3).

Proof. $V_i C / (K_i^M + C) \geq q_i^0$ rearranges to $C \geq K_i^M q_i^0 / (V_i - q_i^0) = C_i^*$. The pool level falls monotonically with supply, so parts cross their thresholds in descending order of C_i^* . With η_i equal, $C_i^* \propto K_i^M$. With thresholds far apart, at most one part is near its threshold at any C (close to strict); otherwise several are short together (shared). ■

S1.8 Rate decides harm (Proposition 6 in version 1). For a renewable part with basal maintenance met and its route back intact, let the units its access can renew fall by φ per step to a depth Δ .

1. If $\varphi \leq \theta K$, no units are lost at any depth.
2. If $\varphi > \theta K$, the loss satisfies $\Delta(1 - \theta K / \varphi) - \tau_f(\varphi - \theta K) \leq \Lambda \leq \Delta(1 - \theta K / \varphi)$.
3. A scar is possible only if $\Delta(1 - \theta K / \varphi) > Q^* K$, and certain if the lower bound exceeds $Q^* K$.

Depth without speed never harms; speed without depth never scars; a scar needs both, and the threshold depth falls as speed rises (Figure S1). Fixed capital ($Q^* = 0$) is scarred by any fall faster than its switch-off rate.

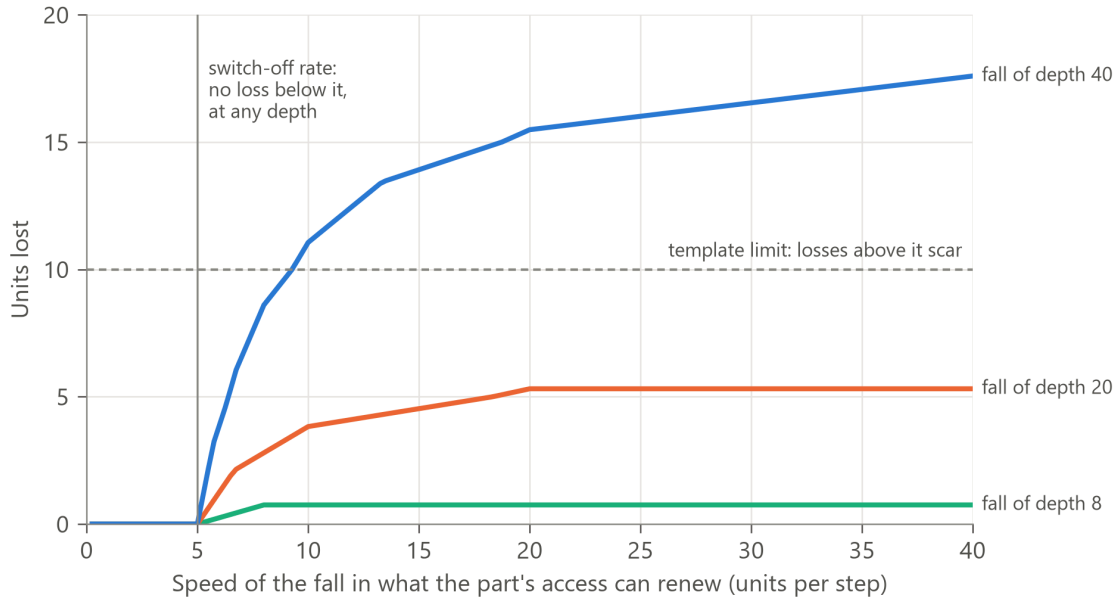


Figure S1. Rate decides harm (S1.8), simulated. A part of $K = 100$ units, with its upkeep met and its route back intact, switch-off rate $\theta K = 5$ units a step and failure time $\tau_f = 4$ steps, faces a fall in the units its access can renew, of depth Δ (8, 20 or 40) and speed φ . Below the switch-off rate no units are lost at any depth. Above it, losses rise with speed, and further the deeper the fall; only the deepest fall crosses the template limit $Q^* K = 10$, beyond which losses scar. Parameter values are illustrative. (Figure 4 in version 1.)

Proof. Over the episode, switched-off and lost units together number Δ . During the decline, which lasts at least Δ/φ steps, the excess x of active over renewable units satisfies $x_{t+1} = (x_t - \theta K)(1 - 1/\tau_f) + \varphi \geq \varphi > \theta K$, so θK units are switched off each step and at least $\theta K \Delta/\varphi$ in all: hence the upper bound. At the end of the decline the excess is at most $\theta K + \tau_f(\varphi - \theta K)$, and no more can be switched off in the tail: hence the lower bound. If $\varphi \leq \theta K$, every unit not renewed is switched off within the step. ■

S1.9 Exhaustion, severance and constriction (Proposition 7 in version 1). Let delivery to the top and its supports be bounded by pathway capacity, the minimum cut between source and part (Ford and Fulkerson 1956). If, in a step in which the support parts were met in the step before and no unit of the top or a support part is still coming back from an earlier shortfall, the protected flow falls while a part outside the top and its supports still draws anything, or a store still has release headroom, then either the minimum cut to the top or a support part is below its need (severance at zero, constriction above zero), or damage from outside has removed units of the top or a support part.

Failure while a part outside the top and its supports is still supplied, with the supports met in the step before and no unit of the top or a support part still coming back, identifies a cut or narrowed link, or outside damage. Each is observable independently. Where the supports were short in the step before, the top can be short while every other part draws in full: that is the dependency lag, counted at the top, not a break in the order. The earlier support shortfall must be observed, not inferred from the later one. Where units of the top or a support part, switched off or lost in an earlier shortfall, are still coming back, the top can be short while every other part draws in full: that is the recovery lag. It is not a break in the order, and the earlier shortfall must be observed, not inferred from the later one (worked example in S2).

Proof. If a part outside the top and its supports still draws anything, then $S \geq N + P$, since the version 2 order serves the top and then the supports in full before any other part; a store with headroom releases the whole gap, so again $S \geq N + P$. With the supports met in the step before, delivery to the top is not limited by the dependency. So the flow could meet the top and its supports. If they are short, another bound binds: the pathway, whose capacity is the minimum cut, or their own capacity, which falls only through an earlier shortfall, whose units would still be coming back (excluded), or through damage. ■

The dependency lag, a worked example. The top needs 10; one support part draws 3 (upkeep 1, work 2); three other parts draw 3 each. Delivery to the top in a step is in proportion to the support part's work in the step before. In step 1, with the support met in step 0, supply is 11: the top receives 10, and the support 1, which covers its upkeep and leaves no work; the other parts receive nothing. In step 2 supply is 22, enough for every draw: the support and the other parts draw in full, but delivery to the top is $10 \times 0/2 = 0$. The top is short while every other part draws in full, with no cut and no damage. The cause is the support shortfall in step 1, which must be observed (S2 checks this case and the restated results).

S1.10 Rerouting (Proposition 8 in version 1). A pathway with parallel routes loses route e . If its flow is reallocated where there is headroom, no surviving route is overloaded if and only if the surviving headroom covers the lost flow. If flow divides by a physical rule (in proportion to conductance), a surviving route can be overloaded even when total headroom suffices. Where a collateral joins two parts' branches, flow reaching one through it is taken from the other at fixed supply (steal).

Cascades and steals are expected in physically divided networks (vessels, pipes, power lines) even with spare capacity, and in networks where flow is placed by access settings (budgets, routers) only when total capacity is short. A budget moved to a channel with room is an instance of the second.

Proof. Placing the lost flow into surviving headroom succeeds exactly when the headroom suffices; in a general network this is max-flow min-cut. By example: routes with capacities 10, 4 and 10 carry 5, 3.9 and 5. Losing the first leaves headroom 5.1 for flow 5, but a split in proportion to capacity sends about 1.43 to the second route, which then carries 5.33 against capacity 4. At fixed supply, conservation at a junction means flow leaving one branch through a collateral is subtracted from it. ■

S1.11 Collapse, not death (Proposition 9 in version 1). A system outside its viable set is inside the capture basin (Aubin 1991), so in collapse rather than death, if:

- supply, including outside input, can return to and be held above the top's need, every surviving part's basal maintenance and, where the top depends on them, the support parts' draws, with a positive remainder for rebuilding;
- the top is not scarred below what the protected flow requires;
- every non-bypassable link the protected flow depends on has capacity above zero or can be restored;
- every part the protected flow depends on has its template intact and its route back open;
- any part whose work produces a resource that rebuilding needs (for example, bone marrow) has positive capacity, or can itself be restored.

The time to re-enter the viable set is at least the longest rebuild time among those parts. The conditions are measurable before the outcome.

Proof. Under the first condition an access setting exists that fully funds the top and its supports and gives every other surviving part access equal to its upkeep, which what reaches it covers first, so the top and its supports lose no further units (Proposition 4 and S1.8). Other parts may still lose units, but the protected flow does not depend on them. The positive remainder funds rebuilding, and under the remaining conditions each needed part is rebuilt at its rebuild rate. The system therefore reaches the viable set in finite time and can stay there: membership of the capture basin. ■

S1.12 Recovery order and partial refill (Proposition 10 in version 1). The intake is a support part, so after refeeding its draw is served before every part outside the supports; its switched-off and lost units, like every part's, come back from what is left by marginal value. With Ψ the distribution of past episode depths, c_S the cost per unit of a store being short, ν the expected frequency of shortfall and ω the best competing value of a part's binding units, the store refills to $L^r = \Psi^{-1}(1 - \omega/(c_S\nu))$ if $\omega < c_S\nu$, and not at all otherwise.

While parts still bind, a store refills only part way, to L^r , which is higher the more frequent (ν) and the deeper (Ψ) past shortfalls have been; where $\omega \geq c_S\nu$ it does not refill at all. The refill level is the newsvendor critical fractile; the allocation rule is marginal analysis of spares (Sherbrooke 1968).

Proof. The intake is a support part, drawn in step 2 of the ordered draw, before every part outside the supports; reactivation and rebuilding are paid from what is left, by marginal value (main text, Section 3.3, step 4). A store's marginal value $c_S\nu(1 - \Psi(L))$ is non-increasing in L ; greedy allocation by marginal value stops refilling where it falls to ω , the stated quantile. Greedy allocation is optimal for separable concave value (Ibaraki and Katoh 1988). ■

S1.13 Repair under strict rank (Proposition 11 in version 1). This is the ordered draw with a limited repair resource as the flow. Let the repair resource be W per step and binding, and let damage raise the requirement of parts by totals δ_j , indexed in rank order; damaged parts draw on the resource by rank. Part i finishes healing at step $\lceil \sum_{j \leq i} \delta_j / W \rceil$. The highest-ranked damaged part heals as fast as alone; each lower-ranked part is delayed by the damage ranked above it. A smaller W never shortens a finishing time, and can lengthen it.

Strict-rank repair is distinguishable from shared repair: under strict rank, the highest-ranked damaged part does not slow when others are damaged too.

Proof. Under strict priority, the cumulative repair resource drawn after t steps is tW , applied to damage in rank order; part i is complete when $tW \geq \sum_{j \leq i} \delta_j$. ■

S2. Numerical checks

What was checked.

- Every quantitative statement in the main text, and every result in S1 except S1.5, S1.6 and S1.11, was checked numerically against an independent implementation of the reduced form (Section 3.3 of the main text).
- The implementation was written from the equations, not from the simulation engine used during the model's development.

- It covers one resource:
 - one step of the ordered draw, in version 2's order: the top's need, then the support parts' full draws, then every other part's full draw, each in rank order, with upkeep before work inside each part (version 1's order and the alternative considered in version 2, below, are kept as records);
 - the two store-release forms;
 - unit switch-off and failure;
 - saturable uptake;
 - strict-rank repair.

Design. Each check draws random instances from stated ranges and tests the proposition's statement on every instance. A single failure prints the counter-example. The seed is fixed (20261007), so every run is identical. The code uses only the Python standard library.

Result: every check passes, and the alternative order fails the version 2 checks, as it should.

Result	What is tested	Instances	Outcome
Proposition 1, scarcity regime	The ledger closes (gap = store draw + total shortfall). Total shortfall is unchanged when the support parts and the other parts are randomly re-ranked, and under the alternative order, at fixed supply and store draw	2,000 (one top, two support parts and six other parts, each with upkeep and work; supply uniform up to total need)	Pass
Proposition 1, general identity	$\sum \ell = \Gamma - \sum d - I + R_{\text{unused}} + X$ with gates closed at random (probability 0.3) and allocation up to 1.5 times reference	2,000 (five parts)	Pass
Proposition 2	The top is met if and only if $S \geq N$; the supports if and only if $S \geq N + P$	2,000	Pass
Proposition 2, with outside input	In terms of the gap, with outside input I drawn at random: the supports are met if and only if $\Gamma \leq \sum d + I + D$, and the top if and only if $\Gamma \leq \sum d + I + P + D$	2,000	Pass
Proposition 3	Without the delivery dependency: under proportional release the store left at the break lies between $(1 - k)L^*$ and L^* , with $L^* = (\Gamma - M)/k$. Under full release, less than one step's gap is left. A faster gap leaves more store unused (gaps 1.5, 2, 3 and 4). A gap at or below the margin never breaks	500 (L_0 50 to 200; k 0.01 to 0.2; M 0 to 5; gap between $M + 0.1$ and kL_0), plus fixed cases	Pass
Proposition 3, S1.1 and G27, by simulation	The full draw with one store, half the cases with the delivery dependency ($M = D$) and half without ($M = P + D$), measured to the step at which the margin is used up (with the dependency, the first step the support parts go short; the protected flow breaks one step later): store left then between $(1 - k)L^*$ and L^* ; lead time within one step of the formula; nothing short while the store has headroom; the first part not fully supplied never moves down the order; the top and the support parts stay met until the margin is used up	400	Pass
Proposition 4	Parts short of their draw form a lower segment of the whole rank order (top, supports, others); no part is short while a lower-ranked part draws anything; within a short part, no work is drawn while upkeep is short	3,000	Pass

Result	What is tested	Instances	Outcome
Propositions 2 and 4 and S1.9, across steps, and the dependency lag	Runs of six steps with random supply (some steps at full supply), and delivery to the top in proportion to the support parts' work in the step before; what the cap holds back is either not delivered or left in the flow. In every step whose supports were met in the step before: the top is met if and only if $S \geq N$, the order holds (Proposition 4), and the top is never short while a part outside the top and its supports draws (S1.9, with no cut or damage). The dependency lag must occur, and the worked example in S1.9 is reproduced	3,000 runs, both variants; 2,547 lag steps found	Pass
Propositions 2 and 4 and S1.9, across steps, with unit states, and the recovery lag	The top and one support part made of units: a unit whose work goes unmet is switched off (up to a set number a step), takes upkeep only and does no work, and comes back at up to a set number a step, paid from what is left; random supply over ten steps, with and without the delivery dependency. In every step in which the support parts were met in the step before and no unit of the top or a support part is still coming back: the top is met if and only if $S \geq N$, and the top and the supports are never short while a part outside them draws. Otherwise the recovery lag must occur. The worked example (from the final check): a top of 10 units (upkeep 0.2 and work 0.8 each), one support part of 3 (upkeep 1, work 2) and three other parts of 3, switch-off 5 units a step and reactivation 1; supply 22, then 6, then 22: the protected flow is 6.8, 7.6, 8.4 and 9.2 of 10 in steps 3 to 6, with the support met in the step before and every other part met in full	3,000 instances, each with and without the dependency; 210 recovery-lag steps found	Pass
The alternative order, fed to the Proposition 4 and G27 checks	The order that keeps every other part's upkeep ahead of any other part's work (considered and not adopted in version 2): the checks must fail	3,000 and 100	Fail, as required (1,719 and 100 failures)
S1.7 (was Proposition 5), strict limit	With four parts of equal capacity and half-saturation constants spaced by factors $R = 10^2, 10^4, 10^6$, saturable uptake departs from strict priority by at most 0.15, 0.02 and 0.002 at five supply levels. Shares fall with the half-saturation constant	Fixed cases	Pass
S1.7 (was Proposition 5), adequacy order	Parts lose adequate supply in descending order of $C^* = K^M \eta / (1 - \eta)$. Counter-example to affinity alone: $V = (1, 10)$, $K^M = (1, 10)$, $q^0 = (0.9, 0.5)$; the higher-affinity part loses adequate supply first	200 random three-part cases (near-ties, with $ \ln(C_i^*/C_j^*) < 0.05$, skipped), plus the counter-example	Pass
S1.8 (was Proposition 6)	A fall in renewable capacity no faster than θK loses no units. A faster fall loses between $\Delta(1 - \theta K/\varphi) - \tau_f(\varphi - \theta K)$ and $\Delta(1 - \theta K/\varphi)$	400 ($K = 100$; θ 0.02 to 0.2; τ_f 2 to 10; φ 0.2 to 40; Δ 5 to 90)	Pass
S1.10 (was Proposition 8)	When a route is lost, reallocating its flow into headroom overloads a surviving route if and only if total headroom is less than the lost flow. A split in proportion to capacity overloads a route in at least one case with enough headroom	2,000 networks (three to six routes, capacities 1 to 10)	Pass
S1.12 (was Proposition 10)	Refilling a store unit by unit while its marginal value exceeds the best competing value ω stops within 0.2 of the critical fractile $\Psi^{-1}(1 - \omega/(c_S \nu))$. There is no refill when $\omega \geq c_S \nu$	500 past episode depths (exponential, mean 20); $c_S = 10$, $\nu = 0.4$; five values of ω	Pass
S1.13 (was Proposition 11)	Under strict rank each damaged part finishes at step $\lceil \sum_{j \leq i} \delta_j / W \rceil$. The highest-ranked damaged part heals as fast as alone	1,000 (five parts; W 0.5 to 5; damage 0 to 10)	Pass
S1.1	Without the delivery dependency, the G27 window, from the store's headroom running out to the break, is within one step of $\ln(\Gamma/(\Gamma - M))/(-\ln(1 - k))$	500 (ranges as for Proposition 3)	Pass

Result	What is tested	Instances	Outcome
S1.2	Allocating each of two complementary resources by its own conflicting order gives no work. Every common split gives total work 1	Exact case	Pass
S1.3	After shrinking, the remaining units are exactly funded, and the loss is no larger than the unfunded share	2,000	Pass
S1.4	Economising at any speed leaves renewal fully funded. Its first-step saving equals $(\kappa^w + \kappa^u)w^0\epsilon$. Consolidation brings active units down to the economised work	300 (θ 0.02 to 0.2; economising depth 0.1 to 0.9; speed one to ten steps)	Pass

Not checked numerically. These follow directly from the definitions, so a numerical check would only restate them:

- S1.11 (was Proposition 9), a constructive condition for re-entering the viable set;
- S1.5 and S1.6.

The code also checks one result that is not used in this paper: a drain treated as a cut in supply. It belongs to the proposed nested-systems extension.

Corrections the checks and review made. Each was confirmed numerically after it was found, except item 6 (S1.11), which is not checked numerically (above):

1. **S1.8 (then Proposition 6):** the loss expression $\Delta(1 - \theta K/\varphi)$ was first stated as an estimate. The checks showed it is an upper bound, and the lower bound was added.
2. **Economising:** an earlier statement implied that fast economising causes unit loss. The checks (S1.4) showed it does not: economising cuts work, not renewal.
3. **Proposition 1:** total shortfall was first stated as invariant to access in general. Review showed this holds only where the flow is fully used and no part receives above its reference. The proposition now states two regimes, and the general identity (with $+X$, not $-X$, as first written) is checked.
4. **S1.7 (then Proposition 5):** rank under saturable uptake was first attributed to affinity alone. Review showed it depends on capacity and requirement as well, and the counter-example above is now part of the check.
5. **Proposition 2 (outside review):** the silence condition left out outside input. It is now $\Gamma \leq \sum_s d_s + I + M$. With random outside input, the earlier condition gave the wrong answer in 622 of 2,000 cases; the corrected one passes all 2,000 (table above).
6. **S1.11 (then Proposition 9; outside review):** the first condition funded the top and basal maintenance but not the support parts the top depends on, nor the repair parts, so it did not guarantee recovery. The condition now requires both, with a positive remainder for repair. A condition that the repair parts have capacity, or can be restored, was added, and the proof was revised. (Version 2 withdraws repair parts: the remainder funds rebuilding, and the capacity condition applies to any part whose work produces a resource that rebuilding needs.)
7. **Phase order (version 2):** changed to rank first, part by part (the top, then the support parts in full, then every other part in full, in rank order; upkeep before work inside each part). Propositions 2 to 4, S1.1 and G27 were re-derived and rechecked; version 1's order, and the alternative considered, are kept in the code as records.

8. **Propositions 2 and 4 and S1.9 (final check, version 2):** the statements failed when a support part was partly supplied, or short in the step before. They are restated for steps in which the support parts were met in the step before, with the dependency lag named, and checked (table above).
9. **Propositions 2 and 4 and S1.9 (final check, version 2, second round):** the statements failed while units of the top or a support part were still coming back after an earlier shortfall. The guard now excludes such steps, the recovery lag is named, and the check gains unit states (table above).

Faults in the checks themselves, fixed before the results above:

- the generator for Proposition 3 produced gaps below the margin, which cannot break, so those cases are now excluded by construction;
- the uptake solver's search range was too narrow for widely spaced half-saturation constants, so it now bisects on the logarithm of the pool level;
- one check for rising requirement tested nothing and was removed.

Code and output.

- Code: `scripts/pam_propositions_check.py`.
- Output: `scripts/pam_propositions_check_output.txt` (every check "ok"; "ALL OK"). The version 2 checks are `check_P1_v2b`, `check_P2_v2b`, `check_P4_v2b`, `check_P3_S11_G27_v2b` and `check_lag_v2b`; the checks with working labels P1 to P18 below are version 1's, kept as its record.
- To run: `python scripts/pam_propositions_check.py`.
- The script's own comments use working labels P1 to P18:

Paper result	Working label
Proposition 1	P1
Proposition 2	P2
Proposition 3	P4
Proposition 4	P3
S1.7 (was Proposition 5)	P17
S1.8 (was Proposition 6)	P6
S1.9 (was Proposition 7)	P8
S1.10 (was Proposition 8)	P13
S1.11 (was Proposition 9)	P15
S1.12 (was Proposition 10)	P9
S1.13 (was Proposition 11)	P10
S1.1	P5
S1.2	P16
S1.3	P7
S1.4	P11
S1.5	P12
S1.6	P14

S3. Prior theories against the model's features

Each row is a theory or family of models; each column is a feature of the model (main text, Section 3); only the seven components of Section 3.1 are called components. Table 1 of the main text condenses this table.

Codes:

- **F**: present in formal (mathematical) form.
- **W**: stated in words.
- **E**: measured, without a general model.
- **p**: partial, or a different form (see the note).
- **·**: not found in what was read.
- **?**: not checked.
- **T**: a pre-registered test result (Tst column only).
- **†**: the row rests on abstracts only and is provisional.
- **Bold** (This model row): no other row has an F in that column (for Tst, a T).

How much of each source was read is listed after the notes.

Columns:

Column	Feature
Gov	A governor that regulates access, with modes and gates
Acc	Order of loss set by access (transport, gating, affinity, pathway), per resource
Pre	That order fixed beforehand from documented access, then tested
Sto	Stores drawn first; a protected floor; intake held or raised
Uni	Units active, switched off, lost or scarred
Rep	Repair competing for access
Led	A conserved ledger of shortfall: unmet requirement counted at each part, which no setting of access removes outside the named exceptions (Proposition 1), including what the protected flow does not show
Via	Viability: collapse against death; a threshold switch before passive failure
Scr	Scars and the template (lasting residue; replacement needs a template)
Res	More than one resource
Sys	Systems other than bodies
Tst	Pre-registered tests

Theory	Gov	Acc	Pre	Sto	Uni	Rep	Led	Via	Scr	Res	Sys	Tst
Dynamic Energy Budget theory (Kooijman 2010)	p ¹	p ²	·	F	·	· ³	p ⁴	F ⁵	p ⁶	F	· ⁷	·
Selfish Brain (Peters et al. 2004; Peters and Langemann 2009; Göbel and Langemann 2011; Peters, McEwen and Friston 2017)	F	F ⁸	p ⁹	F	·	·	p ¹⁰	·	·	·	·	T ¹¹
Energetic model of allostatic load (Bobbá-Alves, Juster and Picard 2022)	W	W ¹²	·	W	·	W ⁵⁵	W ¹³	·	W	·	· ¹⁴	·
Brain-body energy conservation model (Shaulson, Cohen and Picard 2024) [†]	W	W	·	?	·	?	?	?	W	·	·	·

Theory	Gov	Acc	Pre	Sto	Uni	Rep	Led	Via	Scr	Res	Sys	Tst
Plant hierarchical models (Grossman and DeJong 1994; reviewed by Lacoine 2000 and Marcelis and Heuvelink 2007)	.	. ¹⁵	.	p	p ¹⁶	.	p ⁴	.	.	p	.	.
Plant transport-resistance models (Thornley 1972; Minchin, Thorpe and Farrar 1993; Minchin and Lacoine 2005) [†]	.	F ¹⁷	.	p	.	.	p ⁴	.	.	p	.	.
Functional equilibrium in plants (as reviewed by Marcelis and Heuvelink 2007)	p ¹⁸	.	.	F ¹⁹	.	.	.	.	.	F	.	.
Allotaxis (Sterling 2012, 2018; Schulkin and Sterling 2019)	W	. ²⁰	.	W	.	W ²¹	W ²²	.	W ²³	W	W ²⁴	.
Central governor (Noakes 2000, 2012)	W	.	.	W ²⁵	W ²⁶	.	.	W ²⁷	.	W	.	.
Selfish immune system (Straub 2014)	W ²⁸	W ²⁹	.	W ³⁰	.	W	W ³¹	W ³²	W	.	.	.
Hypoxia tolerance (Hochachka et al. 1996) [†]	W	.	.	?	W ³³	W ³⁴	.	W ³⁵	.	.	.	.
Triage theory (Ames 2006; McCann and Ames 2009, 2011)	. ³⁶	W ³⁷	p ³⁸	W ³⁹	p	W ⁴⁰	W ⁴¹	.	W	W ⁴²	.	.
Hierarchy of ATP consumers (Buttgereit and Brand 1995) [†]	. ⁴³	E/F ⁴⁴	.	.	.	E ⁴⁵	.	.	.	.	.	.
Haemodynamic models (Guyton, Coleman and Granger 1972; Hester et al. 2011; Barnes et al. 2020; Curcio et al. 2020) [†]	F ⁴⁶	F ⁴⁷	.	F ⁴⁸	.	.	p ⁴	p ⁴⁹	.	p	.	.
Fetal brain-sparing models (Luria et al. 2012; Garcia-Canadilla et al. 2014) [†]	p	F	.	.	.	.	.	.	.	p	.	.
Maintenance-Growth Model (Mauritsson and Jonsson 2023)	.	.	.	F ⁵³	.	F ⁵⁴	p ⁴	.	.	.	.	.
Disposable soma (Kirkwood 1977; Drenos and Kirkwood 2005) [†]	.	. ⁵⁰	.	.	.	F ⁵⁰	.	p	p	.	.	.
Ultrastability (Ashby 1952)	p ⁵⁶	.	.	.	.	.	.	W ⁵⁷	.	.	p ⁵⁸	.
Perceptual control theory (Powers 1973) [†]	F ⁵¹	.	.	.	.	.	.	.	.	.	?	.
Free energy principle (Friston 2010, 2013) [†]	p ⁵⁹	.	.	.	.	.	.	p ⁶⁰	.	.	p ⁶¹	.
This model	F	F	F	F	F	F	F	F	F	F	F	T ⁵²

Notes

1. DEB adds control species by species; the κ rule is fixed, not governed.
2. DEB's account of κ through carrier densities, and its synthesising units, are access-like at the receiving end.
3. DEB has no repair process: damage is irreparable. Defence sits in maturity maintenance, which is "more facultative" than somatic maintenance (cut first).
4. Mass and energy (or carbon, or fluid volume) are conserved, but there is no account of which parts go short.
5. Death at a structural floor; hazard from ageing.
6. Ageing damage accumulates; replacing damaged cells needs undifferentiated cells (the model's template), stated but not modelled as a scar rule.
7. All organisms, but not organisations.
8. Insulin-gated access: insulin-dependent (GLUT4) against insulin-independent (GLUT1) uptake, in two compartments.

9. Predictions are fixed beforehand and set against a rival theory, but the order is the theory's premise, not derived from access resource by resource. Each model term is assigned one documented mechanism (Peters and Langemann 2009, Table 1), close to the model's mapping step.
10. The supply-chain model is formal and energy-conserving: unmet flow "propagates retrograde" and builds up in front of the bottleneck (fat, blood glucose). Sprengell, Kubera and Peters (2021b) read post-stroke hyperglycaemia as the brain's own pull on supply. Supplement S10.1 sets out why the related trial is not support. Peters, McEwen and Friston (2017) state in words that, if the brain cannot reduce uncertainty, a persistent cerebral energy crisis may develop that burdens the individual as allostatic load.
11. Pre-registered systematic reviews (PROSPERO) of the theory's predictions against a rival theory (Sprengell, Kubera and Peters 2021a, 2021b).
12. "The most urgent processes divert or steal energy from less urgent ones": urgency, judged after the fact.
13. The cut to growth, maintenance and repair can occur "without elevating total energy expenditure", visible only in molecular sequelae.
14. The same process at whole-body, cellular and mitochondrial levels: nested levels, not other kinds of system.
15. Strict priority among organ groups, but the sequence is assigned, not derived from access.
16. Fruit abortion at a low source-sink ratio, the nearest thing to a loss rule.
17. "Sink priority being an emergent property of the model" (Minchin and Lacombe 2005), arising from transport resistance and sink kinetics.
18. Teleonomic: the plant behaves as if seeking a root-shoot balance.
19. Shortage of a resource shifts allocation to the organ that takes it in, as the model's intake is held.
20. "To each organ according to its need": priority set by the brain's prediction of need, not by access.
21. An anabolic (grow and repair) mode set by the circadian clock.
22. Sterling describes treated hypertension as each blocked route moving the load to the next, until the cost lands on exercise (Sterling 2018); the wording is his.
23. Arteries thicken and stiffen, and the system loses the ability to return to normal pressure.
24. Society and the planet as regulated systems: an essay argument, not a mapping.
25. Exercise always ends with a reserve: 35 to 60% of muscle recruited.
26. Motor units are recruited and de-recruited.
27. Exercise is stopped "before there is a catastrophic failure of homeostasis": an active switch before passive failure.
28. Two co-equal regulators, brain and immune system, each able to take control and inhibit the other. In this model they are two modes of one governor function, selected by signals; which mode wins in conflict is an open question.
29. Insulin dependence decides which organs gain from insulin resistance.
30. A non-negotiable floor (about 8,500 kJ a day) and a negotiable remainder; stores last 19 to 43 days.
31. Energy arithmetic of the reallocation (kJ a day), not a conserved account of unmet requirement.
32. Acute programmes must end within the store's time; chronic activation is "a misguided acute program".
33. Channel arrest, spike arrest and translational arrest.

34. Protein synthesis, one of the cell's ATP consumers, is cut (translational arrest).
35. Arrest is reversible in tolerant cells and "irreversible" in sensitive cells.
36. No central regulator: the mechanism is distributed (binding affinities, isozymes, distribution).
37. Binding affinity, transporters, preferential distribution, and a selenium-sensitive tRNA.
38. Essentiality is classified first, from knockout lethality, then checked against the order of loss (vitamin K-dependent proteins, selenoproteins). The classification is not by access and was not pre-registered.
39. Transport proteins are induced first ("homeostatic adjustments"), then triage follows.
40. DNA-repair enzymes lose to enzymes of ATP synthesis.
41. "Insidious changes accumulate" while critical functions stay intact: the cost does not show in them.
42. About 40 micronutrients; worked through for vitamin K, selenium and iron.
43. Control is "widely shared"; no block has more than a third of it.
44. A measured order (macromolecule synthesis, then sodium cycling, then calcium cycling, then proton leak), analysed by metabolic control analysis.
45. Among the cell's ATP consumers, protein and RNA/DNA synthesis are the most sensitive to supply.
46. Baroreflex and other control loops in formal models.
47. The regional order emerges from bed resistances, reflex gains and autoregulation, calibrated to data.
48. Venous unstressed volume is mobilised as a store.
49. Time to cardiovascular collapse.
50. Repair investment is set by optimisation (fitness), not by access.
51. Hierarchical control: higher levels set the reference levels of lower ones.
52. PT1 (main text, Section 5.1): pre-registered and adjudicated blind; one supported result, at half weight.
53. Non-negotiable basal maintenance and feeding costs are paid first; negotiable maintenance and growth share the rest.
54. Negotiable maintenance (defence: immune system, buffering) is cut as a power of relative food intake.
55. Repair competes with growth and maintenance for the energy that allostasis leaves, stated in words; no order among them is stated.
56. Essential variables are held within physiological limits by adaptive behaviour (Sections 3/14 and 5/3): targets on sensed levels, not a regulator of access.
57. Survival is staying within a region of the phase space (Section 5/9). Step-functions change value at critical states nearer the normal values than the limits (Sections 9/1 and 9/5): a switch before failure, stated mostly in words in the sections read.
58. The homeostat, a machine built to the definition of the ultrastable system (Section 8/8), as well as the organism.
59. Internal states appear to minimise free energy (surprise) through perception and action: regulation of the system's own states, with no shared resource or ranked access.
60. Persistence as preserving functional and structural integrity, with homeostasis as a consequence (Friston 2013).

61. Any (ergodic) random dynamical system with a Markov blanket, by a heuristic proof with simulations that "suggests" it (Friston 2013); read as an abstract only.

What the table shows.

- **Every column is filled by some prior theory, most often in words or in part;** no prior theory has Pre, Uni, Led, Scr or Sys in formal form. The model's governor, access-based order, stores, units, repair competing for access and scars each appear elsewhere.
- **No prior theory has more than three columns in formal form,** and the fullest row (triage theory) fills eight of twelve, mostly in words or in part. The nearest are:
 - the Selfish Brain: formal and tested, but one resource and two compartments;
 - the selfish immune system: in words, energy only, two regulators;
 - triage theory: in words, many resources, several levels.
- **Five columns have no formal entry in any prior theory** (Table 1 shows three of them):
 - **Pre:** a rank fixed beforehand from documented access and then tested. Triage theory (by essentiality, not access) and the Selfish Brain (as a premise, not derived) come closest.
 - **Uni:** units active, switched off, lost or scarred. Stated in words for the central governor and hypoxia tolerance, and in part in plant hierarchical models and triage theory.
 - **Led:** a conserved account of which parts go short. It is stated in words by the energetic model of allostatic load, by allostasis, by the selfish immune system and by triage theory, but never kept as an account.
 - **Scr:** scars and the template. Stated in words by the energetic model of allostatic load, the brain-body energy conservation model, allostasis, the selfish immune system and triage theory, and in part by DEB and disposable soma.
 - **Sys:** the same rules applied outside bodies. Allostasis argues it in words, as an essay (note 24); the energetic model of allostatic load spans nested levels, not other kinds of system (note 14). The free energy principle suggests, by a heuristic proof, that its own principle holds for any (ergodic) random dynamical system with a Markov blanket (note 61), but it allocates no shared resource.

How much of each source was read:

- **In full:**
 - Bobba-Alves, Juster and Picard 2022; Peters and Langemann 2009;
 - Sprengell, Kubera and Peters 2021a and 2021b; Sterling 2018;
 - Noakes 2012; Straub 2014; Ames 2006; Mauritsson and Jonsson 2023;
 - Marcelis and Heuvelink 2007 (for the plant model classes).
- **Sections read:** Kooijman 2010; Ashby 1952 (Sections 3/14, 5/3, 5/9, 7/1, 8/4, 8/8, 9/1 and 9/5, in the 1954 reprint).

- **Abstracts only:**
 - Peters et al. 2004; Göbel and Langemann 2011; Shaulson, Cohen and Picard 2024;
 - Lacoïnte 2000; Minchin and Lacoïnte 2005;
 - Sterling 2012; Schulkin and Sterling 2019; Noakes 2000;
 - Hochachka et al. 1996; McCann and Ames 2009 and 2011; Buttgerit and Brand 1995;
 - Barnes et al. 2020; Curcio et al. 2020; Garcia-Canadilla et al. 2014; Luria et al. 2012;
 - Drenos and Kirkwood 2005; Powers 1973;
 - Friston 2010 and 2013; Peters, McEwen and Friston 2017.
- **Cited for the tradition they founded, not read here:**
 - Thornley 1972; Minchin, Thorpe and Farrar 1993 (title and review accounts);
 - Grossman and DeJong 1994 (via Marcelis and Heuvelink 2007);
 - Guyton, Coleman and Granger 1972; Hester et al. 2011;
 - Kirkwood 1977.

S4. All predictions, with status

The model's predictions are G1 to G27. Their wording here is the paper's; model vo.20, Section 10, lists the same predictions, some more briefly. Main-text Table 2 lists the predictions that follow from Propositions 1 to 4 alone, every prediction that has been tested, and G27, which replaces the tested G12 (its window uses S1.1).

Layers: layer 1 is what any feedback loop that holds a flow by drawing a finite store would be expected to show; layer 2 needs selection or design.

Evidence labels:

- **derived:** follows from the reduced form; the proposition is named;
- **simulation result in a development engine (version 1's order or earlier):** reproduced by an engine used during the model's development (the engines are not part of this record). Simulation results come from the development engines (tq_core, tq_min, tq_alloc and tq_units). These ran version 1's order or earlier orders, each with its own recovery rule, and have not been rerun under version 2's order. Where an engine built a result into its rules (G6, G10, G22), the label is withdrawn;
- **natural-system observation:** consistent with published findings read for the checks in S6, which are not tests;
- **compatible:** the direction fits, but the finding cannot discriminate;
- **direct test:** a pre-registered held-out test (main text, Section 5.1);
- **not yet tested.**

Numbering gaps:

- **G11 was withdrawn** during development: it followed from existing rules and added nothing.
- **G9, G14, G15 and G17 are excluded from this paper.** They were carried from an earlier version without re-examination, so they are listed below as excluded, not as predictions.

No.	Prediction	Layer	Status
G1	The protected flow stays at the top's need while lower parts and stores move. It holds while the gap is no larger than what the stores release, plus outside input, plus what can go unmet below the top	1	Derived (Proposition 2); simulation result in a development engine (version 1's order or earlier). Natural observations of held indicators do not test G1. The one observation of the protected flow (by a proxy, S7.3) fell before the break and is logged against the model. The H1 check (on an indicator) was not consistent (10.7% of 75 cases), at low weight
G2	Lower-ranked parts lose access first, in ascending rank	2	Derived (Proposition 4). Compatible: calibrated haemodynamic models (main text, Table 4). Logged against, not yet weighed: Krieger (1921; S7.2)
G3	With a store meeting the gap and its release not binding, the break comes at the same cumulative gap whatever the rate to within one step's gap (Proposition 3); with a delivery dependency, the cumulative gap at the break is larger by one step's gap, so it rises slightly with the rate. Where release binds (a tapering store, or a gap above the release rate), faster onset breaks earlier. Under proportional release, the margin is used up when the store left is (gap minus margin) divided by the store's turnover rate, so a larger gap per step leaves more of the store unused, rising linearly with the gap; the break comes then, or one step later where delivery to the top depends on the support parts	1	Derived (Proposition 3). Not yet tested. Indicator-defined events in fast and slow haemorrhage in sheep (a 30 mmHg fall in pressure) and fast and slow drought in trees (death) fit the first clause; no break of the protected flow was observed (main text, Table 4)
G4	A larger store gives a longer silence; a depleted start breaks sooner	1	Derived (Proposition 2). Natural observations of indicator-defined events fit (S6); they do not test the protected flow. Logged against, not yet weighed: Hikino et al. (2026; Section 6; S7.2)
G5	A part whose access is cut loses work, then upkeep, and with them units (switched off first, lost if supply falls too fast). Loss begins at the lowest-ranked part; each part above it goes short only once the part below has nothing	1 and 2	Derived (Proposition 4 and S1.8)
G6	After the top, the support parts' draws, the intake's among them, are served before every part outside the supports, and their switched-off and lost units come back by marginal value; the protected flow recovers before the state does	2	Not yet tested. The kidney observation (S6) is of an indicator, creatinine, so it does not test the protected-flow clause
G7	Fixed capital keeps what it loses; renewable parts rebuild lost units, except scarred ones	1	Simulation result in a development engine (version 1's order or earlier); natural-system observation: consistent
G8	Rate decides harm. Renewal falling no faster than a part can switch units off leaves no loss. Faster falls lose units. With the route back intact, a scar needs speed and depth together; the threshold depth falls as speed rises	1	Derived with bounds (S1.8); simulation result in a development engine (version 1's order or earlier). Not yet tested
G10	After a short episode, parts come back before stores; after a long one, stores come first	2	Natural-system observation: consistent. Qualitative unless the costs are mapped (S1.12)
G12	Recovery from small knocks slows before the break when the break is approached through shrinking release headroom (a tapering store, or a gap rising towards the release rate). It does not when the break comes with headroom intact (a store with full release emptying, or a switch)	1	Direct test: failed in H1 (blood loss under anaesthesia; release profile committed as a taper, possibly ending in a switch), at half weight. Runnable only under the pre-data reading of missing bins; the step-down inside the fallback cohort was not stated in the pre-registration. Lead time derived (S1.1). Its derivation put the warning in the protected flow, which Proposition 2 says is silent; H1 measured an indicator. The verdict stands (main text, Section 5.1; G27)

No.	Prediction	Layer	Status
G13	The lowest-ranked part goes short first and is cut fully before the next is touched; it stays short longest where its value in recovery is also lowest (recovery is by marginal value, not rank)	2	Derived (S1.5)
G16	Economising is a governor mode, documented in advance like any other mode (S9.2), in which every part below the top and its supports has its access lowered by the same share of its work draw, so work is cut evenly while what reaches each part still covers its upkeep first. It is not the default order. It causes no unit loss at any speed, because work is cut, not renewal; its saving arrives at once for work and wear, and as units are switched off for their baseline renewal; stores are preserved. The top is kept out of economising; whether a top cuts its own work is left to a later version (S1.4)	2	Derived (S1.4); simulation result in a development engine (version 1's order or earlier)
G18	Parts sharing a dependency that is short move together before the break; parts clustered only by rank do not	1 and 2	Simulation result in a development engine (version 1's order or earlier). Not yet tested
G19	Three outcomes (switched off, lost, scarred) can be told apart by what comes back and how fast	1	Simulation result in a development engine (version 1's order or earlier). Not yet tested
G20	Exhaustion and severance. With routes intact and no outside damage, the support parts met in the step before and no unit of the top or a support part still coming back, no part loses units while a lower-ranked part still draws anything, and the top goes last. Failure while a part outside the top and its supports is still supplied, under the same conditions, means a link cut or constricted below need (pathway capacity as maximum flow), or damage from outside to the top or its supports. Where the supports were short in the step before, the top can be short while every other part draws in full (the dependency lag, S1.9); where units of the top or a support part are still coming back, the same can happen (the recovery lag)	1 and 2	Derived (Proposition 4; S1.9). Not yet tested
G21	The law of the minimum. A part's work falls in proportion to its scarcest resource while other resources are ample; what it cannot use stays in the flow, and surplus is spilled	1	Simulation result in a development engine (version 1's order or earlier). Known (Liebig's law); general form in DEB's synthesising units (Kooijman 2010)
G22	The intake coasts. With nothing to take in, the intake switches its units off on basal maintenance; at refeeding its draw, as a support part, is served before every part outside the supports, and its units come back by marginal value	2	Derived (S1.12)
G23	Repair is work like any other: damage raises the damaged part's requirement, met within the part's rank. (a) Under a sustained shortfall, a damaged part's raised requirement goes short in rank order like any other, so repair slows; this is a consequence, not a rule. (b) Strict rank: with a limited repair resource as the flow, when several parts are damaged at once, those ranked after another damaged part heal more slowly than alone; the highest-ranked damaged part heals as fast as alone. (c) An acute threat leads to store release on a sensed level before damage occurs; this is not a mode	2	(a) Compatible with version 1's statement: psychological stress slowed wound healing (Kiecolt-Glaser et al. 1995; Marucha, Kiecolt-Glaser and Favagehi 1998); not re-weighted against version 2's. (b) Derived (S1.13); not yet tested; it can be told apart from shared repair, in which every damaged part slows. (c) Compatible with version 1's statement: acute stress hormones enhance skin immune function (Dhabhar and McEwen 1999); not re-weighted against version 2's

No.	Prediction	Layer	Status
G24	Cascade along substitutes. Losing or narrowing a route moves its flow onto the rest. Where flow divides by physics (vessels, pipes, power lines), a surviving route can be overloaded, or the displaced flow drawn from another part's branch (steal), even with spare total capacity. Where flow is placed by access settings (budgets, routers), this happens only when total spare capacity is short. Overloaded routes fail in turn, so failure spreads along the substitutes, not by rank	1	Derived (S1.10). Not yet tested as a cross-domain prediction
G25	Order of loss from access. Under scarcity, the order in which parts lose adequate supply is predicted by properties of access documented beforehand (constriction under sympathetic drive, autoregulation, affinity, redundancy; discretionary budgets)	2	Direct test: supported in PT1 (English single-tier councils, 2014-15 to 2019-20, statutory duty classified blind), at half weight: lines with a statutory duty were protected more than discretionary ones. Lines with a duty of uncertain level were not distinguishable from discretionary ones. The scarcity version (G25-C), that the gap widens where funding fell more, was not supported. Compatible: calibrated haemodynamic models (main text, Table 4). Note (version 2): PT1 tested this prediction in its pre-registered form (statutory class predicts the order: A above B above C); where access is built in, the rules for requirement also enter the order (S1.7)
G26	Partial refill. In recovery, a store refills only until its marginal value falls to that of the best competing use, so it need not refill to full while parts are still short. The refill level is the critical fractile $\Psi^{-1}(1 - \omega/(c_S \nu))$ of past episode depths	2	Derived (S1.12). Not yet tested
G27	As the break approaches, fluctuations in the gap show first in the store's release while it has headroom, then in the access of the lowest-ranked part still supplied, then in each part above it in turn, and in the protected flow only at the break. A test of G27 needs a system whose final store tapers	1 and 2	Derived (Propositions 2 and 4; window length S1.1, read in version 2's terms). Not yet tested. Added in version 2; G12 as stated is replaced by G27 (main text, Section 5.1)

Excluded from this paper. These are carried from an earlier version and were not re-examined:

- G9, re-tuning to a past threat;
- G14, peak-referenced protection;
- G15, store memory after a deeper shortfall;
- G17, economising and growth compete.

S5. Held-out test records

Both tests followed the same procedure, in this order:

1. Documentation was gathered by another model, barred from reporting values.
2. A setting map and mapping were written.
3. The pre-registration was frozen.
4. The analysis code was written and tested on synthetic data only.
5. The data were downloaded and counted, then run once to completion with the code unchanged.
6. The computation was replicated by a second AI system, which wrote its own script without seeing the data or the results.

7. A blind adjudication was made against the frozen tables and the adjudication rules (tests/ADJUDICATION_RULES.md), in a separate session with no shared context.

Every unplanned choice is logged as a deviation, with the direction it pushes. Fingerprints (SHA-256) of the frozen files are in the repository README.

S5.1 H1: warning before falls in arterial pressure in surgical cases with heavy blood loss (G12)

Item	Record
Documentation	raw/2026-10-06_gemini_H1-VDB-S1.md (structure only; no values)
Setting map and mapping	tests/H1 VitalDB G12 - setting map and mapping (draft).md
Pre-registration	tests/H1 VitalDB G12 - pre-registration.md, frozen 6 October 2026 (SHA-256 begins 371caf3c)
Model version tested	theory/TIER_QUEUE_MODEL_v0.17.md
Analysis code	tests/scripts/h1_vdb_g12.py (SHA-256 9fbe80d7), unchanged since before the data were opened
Synthetic checks	tests/results/H1-VDB-synthetic/: 7 of 7 verdicts as built. G1 was not challenged by any synthetic case
Count and result	tests/results/H1-VDB/ (counts.json, summary.json, README.md)
Replication	raw/2026-10-06_chatgpt_H1-VDB-R1.md; tests/results/H1-VDB-R1/
Adjudication	raw/2026-10-07_chatgpt_H1-VDB-ADJ1.md

Data and cohorts. VitalDB (Lee et al. 2022).

- **Primary cohort (no vasopressor boluses):** 755 eligible cases; at 20% of estimated blood volume, 0 high-loss and 16 low-loss qualifying cases, both short of the 20 required, so it was not runnable.
- **Fallback cohort:** 2,324 eligible cases. At a loss threshold of 20% of estimated blood volume the high-loss group was short (19 cases). At 15% it gave 25 high-loss and 42 low-loss cases, analysed at half weight as fixed in advance.

Result.

- **P1, the excess rise in lag-1 autocorrelation of mean arterial pressure before falls in high-loss cases:** median -0.072 , one-sided $p = 0.48$.
- **P2, high-loss excess greater than low-loss:** $p = 0.78$.
- **Verdict as computed:** Fails, and not Contradicted (two-sided $p = 0.96$).
- **Sensitivity analyses** (not part of the verdict): Fails in every computable variant.
- **G1 check** (low weight): pressure held while haemoglobin fell in 10.7% of 75 cases, against a pre-stated "more than half". Not consistent.
- **G18** (exploratory, not scored; pre-specified on 6 October under the original framing), as the adjudication recorded it: MAP-HR, median excess change $+0.126$ (25 cases); MAP-SV, median excess change -0.0018 (12 cases).

Replication. The second system's script reproduced the frozen run exactly with both frozen readings of missing 10-second bins (the stable approach and interpolation within a window), with every count, score and test identical; with the frozen stable-approach reading alone, P1's median was

−0.050 ($p = 0.62$), and the verdict was still Fails. Under its own stricter reading (a missing bin fails the stable approach), the test is not runnable (15 high-loss and 31 low-loss cases). The frozen run was kept as the result of record, with this dependency stated.

Adjudication (verbatim): "VERDICT FOR G12 IN THIS SYSTEM: Fails." Weight: "fallback cohort, half weight; 15% EBV high-loss threshold." The adjudicator recorded two qualifications that go with the verdict:

- the test is runnable only under the pre-data reading of missing bins;
- the pre-registration does not state that the step-down to 15% applies inside the fallback cohort, although the code fixed before data applies it.

Deviations: none from the analysis rules or the code. One in procedure: the pre-registration put the blind adjudication before the replication, with the pre-registration and summary.json as the adjudicator's inputs. The replication ran first, and the adjudicator also received its finding on missing bins, so that the verdict could state the dependency. The verdict category under the frozen rules was unchanged.

Correction in version 2 (the verdicts stand).

- **The derivation.** G12's derivation (S1.1) put the warning in the protected flow, which Proposition 2 says is silent before the break. H1 tested G12 as stated, and G12 as stated failed.
- **The mapping.** The corrected mapping redraws the frozen boundary (the patient's circulation plus the anaesthetist) to the patient, with the anaesthetist as an outside loop. At this whole-body boundary the top is the brain, and the protected flow is the oxygen the brain draws from its blood supply, measured against its resting need. Mean arterial pressure is an indicator. It tracks a level the governor holds: the stretch of the arterial wall at the carotid sinus and aortic arch, sensed by the baroreceptors (the arterial baroreflex; Chapleau, Hajduczuk and Abboud 1991). H1 measured an indicator, not the protected flow.
- **The release profile.** The frozen H1 map committed a store that tapers as it empties, possibly ending in a switch. The pre-registration named the release profile as the first candidate for revision if G12 failed, and it is logged as that. Which release profile held in the VitalDB cases is untested. The pre-registration bars explaining the null as a switch. It is not explained, and the verdict stands. A later test under anaesthesia must fix its release profile from evidence under anaesthesia: in Evans et al. (2001), anaesthetic agents blunted or abolished the compensation (for example, halothane) or the switch (for example, alfentanil).
- **G27**, defined and untested: as the break approaches, fluctuations in the gap show first in the store's release while it has headroom, then in the access of the lowest-ranked part still supplied, then in each part above it in turn, and in the protected flow only at the break. A test of G27 needs a system whose final store tapers.

- **G18, examined further** (exploratory, post hoc; tests/scripts/h1_g18_followup.py, tests/results/H1-VDB-G18-followup/; it reproduces the frozen figures exactly).
 - **The measure:** within each 10-minute window, the correlation of detrended mean arterial pressure with detrended heart rate (Solar8000) or stroke volume (Vigileo or EV1000, mL per beat), in 10-second bins, early (31 to 21 minutes before the fall) and late (11 to 1 minutes before), less the same change at control times in the same high-loss cases.
 - **Heart rate:** median correlation 0.16 early and 0.33 late before falls, against 0.20 and 0.12 at control times; it moved away from zero in 72% of cases: heart rate moved with pressure, not against it.
 - **Stroke volume:** median correlation near zero in both windows (−0.07 and −0.01; controls −0.02 and −0.02).
 - **Reading:** G18 as mapped paired two outputs of one part (the heart) with an indicator, not two parts, so it was not a test of G18 as stated. It is not evidence for G27, which concerns the access of parts below the top, and it is not counted.
- **The corrected mapping** is tests/H1 VitalDB G12 - mapping corrected (v2).md, beside the frozen setting map, which remains the test record. It names ten slips in the frozen map.

S5.2 PT1: fixed against dynamic priority, and order of loss from access, in English local government (G25)

Item	Record
Documentation	raw/2026-10-07_gemini_PT1-S1.md, raw/2026-10-07_gemini_PT1-S2.md
Blind classification of statutory duty	raw/2026-10-07_gemini_PT1-C1.md (failed, without files), raw/2026-10-07_chatgpt_PT1-C1b.md, raw/2026-10-07_chatgpt_PT1-C1c.md; frozen as tests/PT1_statutory_classification.csv (176 lines; SHA-256 begins faf45875), with tests/PT1_line_name_crosswalk.csv
Design note, setting map and mapping	tests/PT1 Priority test - design note (draft).md; tests/PT1 Priority test - setting map and mapping (draft).md
Pre-registration	tests/PT1 Priority test - pre-registration.md, frozen 7 October 2026 (SHA-256 begins a4aab81d)
Model version tested	theory/PERSISTENCE_ALLOCATION_MODEL_v0.18.md, as corrected on 7 October 2026
Analysis code	tests/scripts/pt1_councils.py (SHA-256 995abfc5), unchanged throughout; parser tests/scripts/pt1_parse.py
Synthetic checks	tests/results/PT1-synthetic/: 6 of 7 scenarios as built. The seventh was a chance false positive and was correct on reseeding
Procedure log and deviations	tests/PT1 Priority test - procedure log.md
Result	tests/results/PT1/ (counts.json, summary.json, run.log, README.md)
Replication	raw/2026-10-07_chatgpt_PT1-R1.md; tests/results/PT1-R1/
Adjudication	raw/2026-10-07_chatgpt_PT1-ADJ1.md

Data.

- **Sample:** 121 single-tier councils in England, 2014-15 to 2019-20; 93 spending lines; 23,373 rows for PT1 and 18,905 for G25.

- **Access property:** whether a statutory duty attaches to each line, classified blind before any spending data were opened:
 - A: a duty;
 - B: a duty of uncertain level;
 - C: discretionary.

Deviations, each with its direction:

- **D-1:** a name crosswalk: 21 data names in the window years differ from the 2025-26 guidance names the classification uses, and are mapped to them. Decided from names only, before values were seen. Neutral.
- **D-2:** many-to-one lines summed. Neutral.
- **D-3:** a labelling error in the source relabelled by position. Neutral.
- **D-4:** a limitation, with no intervention: allotments are reported separately from 2019-20.
- **D-5:** duplicate codes removed from the funding table after the first run stopped with an error before producing any result. Approved by the author. Neutral. The analysis code was unchanged.

Result.

- **PT1, primary:** $\beta = 0.054$ (SE 0.405), one-sided $p = 0.45$, 95% interval -0.75 to 0.86 .
Inconclusive: the interval includes the smallest effect set in advance as meaningful (0.25).
- **G25 (half weight):** within a council and year, lines with a statutory duty grew in real spending per head about 4.6 percentage points a year faster than discretionary lines (one-sided $p = 2 \times 10^{-11}$). Class B was not distinguishable from C. **Supported.**
- **G25-C (half weight):** the interaction with the depth of the funding fall was 0.22 (SE 0.25). **Not supported.**
- **Sensitivity analyses** (not part of the verdicts):
 - PT1 is inconclusive in every variant;
 - G25's protection of class A holds in every variant;
 - the B-above-C step fails without London.
- **SS (descriptive, not scored):** of 349 council-years in which in-scope real spending fell, 96.8% had some class A line fall while class C lines kept more than half their 2014-15 total. Read as pre-registered, the share is near zero under strict priority and substantial under shared cutting. The measure is coarse: with about a dozen A lines, some A line falls in almost any council-year, so the share says little about strict order on its own (tests/results/PT1/README.md). It is logged in S7.1.

Replication. The second system's script reproduced every count, estimate, standard error, p-value and verdict to at least eight significant figures. One descriptive (unscored) measure, SS, differed because of a definitional choice: 320 council-years and 96.9%, against 349 and 96.8%.

Adjudication (verbatim): "VERDICTS: PT1 Inconclusive (full weight); G25 Supported (half weight); G25-C Not supported (half weight)." No faults were found that make a verdict unsafe. None of the adjudication rules changed a verdict.

Contamination, declared in the pre-registration: G25 carries half weight because the AI collaborator knew the broad pattern in advance.

S6. Natural-system observations

What these are. Twelve surface-level checks, made on 5 October 2026 while the model was being built.

- **Method:** for each, predictions were written down and committed to the repository before any source was opened. Published findings were then read and each prediction was given a verdict.
- **These are not tests**, for three reasons:
 - many outcomes were known in advance to the AI collaborator, and each check declares this;
 - sources were mostly read through search summaries;
 - three early refinements of the model were built from the blood-loss and fasting findings.
- **Weight:** they show that the model is compatible with these systems, nothing more. They carry no weight as support for the model's claims; the sheep result is used only as evidence on a mapping's release profile (main text, Table 4).
- **Full records:** the prediction tables, findings with their sources as read, contamination statements and weights are in `theory/natural_test_*.md`, with the predictions committed in earlier commits than the results.
- **Sourcing:** findings that the main text relies on are cited there from sources read directly (main text, Table 4). Where a later reading corrected a summary-based finding, the correction is noted below.

No.	System	Predictions	Verdicts	Contamination and weight
1	Blood loss and lower-body negative pressure	N1 to N6	3 consistent; N2 partly (lower-priority beds cut first, but brain supply only partly protected); N3 steepness consistent, mechanism different; N6 partly	Class scheme and compensatory reserve known; N4 rests on 8 sheep (Scully et al. 2016)
2	Prolonged fasting	F1 to F7	4 consistent; F6 consistent for starvation but the opposite under food restriction (weak; summary only); F4 and F7 not tested	F1 and F3 known; summaries only, including the threshold source (Cherel and Groscolas 1998)
3	Loss of kidney filtering units	K1 to K6	6 consistent	Summaries only; K1 and K3 known
4	Honeybee colony under worker loss	H1 to H7	5 consistent; H6 and H7 partly	Heavy: most known in advance; H4 at the level of individual bees
5	Plants in drought	P1 to P8	6 consistent; P6 (G4) mostly, with one finding logged against G4, not yet weighed (Hikino et al. 2026; S7.2); P8 not tested	P2, P3 and P5 known. P2 (leaves fail first) was later found contested (Peters and Choat 2025; S7). A figure for embolism in green canopies was not found in its cited source and is withdrawn
6	Fetal growth restriction	FG1 to FG7	7 consistent; FG5 favoured one of two recovery rules	FG3, FG4 and FG6 known; FG5 contaminated
7	What sets the recovery order (two rival rules)	Three discriminating cases	Rule L (type of past load) fitted all three; Rule D (next demand) fitted none as the main driver	Case A contaminated

No.	System	Predictions	Verdicts	Contamination and weight
8	Muscle: acute injury, chronic overload, disuse	M1 to M8	6 consistent; M3 partly (the indicator, maximal performance, was not clearly late in muscle); M4 not tested	Mostly known
9	What counts as a record (five paired cases; "record" is version 1's term, the ratio of the top's output to its level, applied in these checks to the figure observers watch, an indicator in version 2)	R1 to R5	5 consistent	Heavy: all five known in outline; R2's human evidence contested
10	Re-tuning after past threats	RT1 to RT6	5 consistent (RT5 on benefit only); RT6 partly (holds in one group only)	Mostly known in outline
11	Protection referenced to peak load, in muscle	FM1 to FM5	2 consistent; FM2 and FM5 partly; FM3 inconsistent as written	Summaries only; FM1 and FM4 known
12	Weight cycling	WC1 to WC7	2 consistent; 3 partly; WC6 and WC7 not found	Partly known

Checks 6, 10, 11 and 12 bear partly on predictions excluded from this paper (G9, G14, G15; S4).

What recurs across the first five systems (corrected where later reading required):

- A watched figure (an indicator) is held steady while a store is drawn down or a part or route loses capacity:
 - blood pressure in blood loss (an indicator that tracks a level the governor holds: arterial wall stretch, sensed by the baroreceptors);
 - the function of protected organs in fasting;
 - creatinine in kidney loss (not mapped here; not a flow, so not the protected flow);
 - visible brood and stores in a colony losing workers;
 - canopy colour in drought (not mapped here; not a flow, so not the protected flow).
- **State markers show the drawdown or the loss of capacity, and the figure does not:** stroke volume (the heart's work, falling as blood volume is drawn) and compensatory reserve (a store drawn); fat mass (a store drawn); renal functional reserve (spare capacity of a part); adult numbers (units lost); loss of hydraulic conductivity (a route losing capacity).
- **Indicator-defined events come at a set depletion, not at exhaustion, often through an active switch** (none is a measured break of the protected flow):
 - sympathetic withdrawal at about 30% blood loss;
 - phase III of fasting at a threshold fat share;
 - creatinine rising after about half of the filtering units are lost;
 - brood cannibalism when protein runs short;
 - about 80% loss of conductivity in loblolly pine saplings.
- **The starting state sets how much room there is:** heat stress shortened tolerance to blood loss; a leaner start reached phase III sooner; a low starting number of filtering units led to earlier decline.

Two cases moved here from an earlier draft of the main text (illustrations only):

- **Kidney.** Creatinine stays roughly normal until about half of the filtering units are lost; the remaining units filter more, and the renal functional reserve falls first. After acute injury, creatinine returns to baseline while reserve does not. This is from search summaries; K1 was known in advance.
- **Fasting penguins and other birds.** Phase III of a long fast, with protein breakdown, renewed food search and, in breeding birds, egg desertion, begins at a threshold fat share (about 9% in penguins) rather than after a set time. A larger starting store lengthens phase II (Cherel and Groscolas 1998). The model's point at which a store begins to release was built from this source, so this case is a basis, not a check.

Other sources are listed, as read, in each check's file.

S7. Findings logged against the model

Rule. Every finding that does not conform, or may not conform, is logged when it is found, with its source.

- **Weight:** it is weighed by the same standards of data quality and reporting as a supporting finding.
- **Explanations:** an explanation may be offered only if it names the specific part that went short, or the specific fault in the measure or the mapping. It is logged as a prediction before it is checked, and the outcome is logged either way. If the named place or fault is not found, the explanation is struck. A signal fault (S9.5) is named only under the guard there; it is never a rescue.
- **Absence:** "not identified" is not "absent". A null result on the measures a study took is weighed as a result about those measures only.

S7.1 Held-out tests

Finding	Bears on	Weight	Explanation logged in advance
G12 failed in H1: no rise in lag-1 autocorrelation of arterial pressure before falls in high-loss surgical cases (S5.1)	G12	Half weight (fallback cohort), with two qualifications: runnable only under the pre-data reading of missing bins, and the step-down inside the fallback cohort not stated in the pre-registration	Logged in advance (pre-registration): the release-profile mapping as the first candidate for revision if G12 failed, and the bias from bleeding recorded only as a case total (accepted; a null still counts). Version 1 named the same candidate, with the anaesthetist as an outside loop holding pressure; the release profile stays logged and untested. Afterwards (version 2), under the rule for mapping faults: the derivation put the warning in the protected flow, which Proposition 2 says is silent; H1 measured an indicator (mean arterial pressure); a store that tapers as it empties, possibly ending in a switch, was committed; and the corrected mapping redraws the frozen boundary (the patient's circulation plus the anaesthetist) to the patient, with the anaesthetist as an outside loop. Whether these faults caused the failure is untested. Corrected in tests/H1 VitalDB G12 - mapping corrected (v2).md. The verdict stands
G1 check in H1: pressure held while haemoglobin fell in 10.7% of 75 cases, against "more than half"	G1	Low (a check, not a test)	None

Finding	Bears on	Weight	Explanation logged in advance
G25-C not supported in PT1: the protection gap was not detectably wider where funding fell more (interaction 0.22, SE 0.25; S5.2)	G25-C	Half weight	None
PT1's class B (a duty of uncertain level) was not distinguishable from discretionary lines, and the B-above-C order fails without London	G25	Half weight (primary: $\delta_B > 0$ not met, $p = 0.27$; the B-above-C order fails only in sensitivity 7, London excluded)	None
PT1's descriptive measure SS: in 96.8% of 349 council-years with a fall in spending, some class A line fell while class C lines kept more than half their base-year spending; under strict priority the share would be near zero (S5.2)	Proposition 4's strict priority; G13	Low: descriptive, not scored, and coarse (S5.2)	None

S7.2 Literature

Finding	Source	Bears on	Weight and status
After prolonged starvation, heart, liver, pancreas and kidney each lost about 40% of their mass while the brain lost under 2%, with no order among the organs below the brain	Krieger (1921), as cited by Peters and Langemann (2009); original not read	G2 (order of loss below the top)	Not yet weighed. Whether it conflicts depends on the order fixed for energy and protein at a starvation mapping
In 12 Australian tree species, the predicted leaves-before-stems order of hydraulic failure (vulnerability segmentation) was "universally absent or negative"	Peters and Choat (2025), abstract	Natural check 5, prediction P2; not used in the main text	Logged against P2; the leaves-first order is not claimed
In mature spruce, an earlier drought eased physiological stress in a later one, where a stressed start should fare worse	Hikino et al. (2026), abstract	G4	Logged against G4, not yet weighed. Leaf area fell by 60% and was still 30% lower four years later; that does not lower the canopy's requirement, since the reference is never lowered because units were lost (main text, Section 3.4). An earlier explanation by a lowered requirement is withdrawn
Human food insecurity is linked to higher body weight only in adult women in high-income countries	Nettle, Andrews and Bateson (2017), as summarised	Natural check 10 (RT6)	Logged; the model offers no reason for the restriction
Synaptic loss after noise is shown in animals, but one human study with normal audiograms found no evidence of it	Guest et al. (2018), as summarised	Natural check 9 (R2)	Logged as a caveat on that case

S7.3 Natural checks (S6)

Blood loss, mismatch 1: the measure was blood velocity in the middle cerebral artery, a proxy for the brain's blood flow, which tracks the oxygen the brain draws only while the share it extracts is unchanged. It fell by 27% at presyncope while the indicator (arterial pressure) was unchanged, in

nine volunteers under lower-body negative pressure (Bondar et al. 1995). By this proxy, the protected flow was not held before the break.

- **Explanation named at the time, not checked against this source:** the brain's supply depends on cardiac output, which falls in central hypovolaemia (Evans et al. 2001).
- **Outcome:** the mismatch was used to build an earlier version of the model, which let the top share an upstream supply.
- **In version 2's terms** it is the one observation of the protected flow (by a proxy), and it is logged against the model and counts against it (main text, Table 4 and Section 6). The top is fixed by the boundary, so the finding is not a reason to move the top.
- **Logged in version 2:** the same source reports rising cerebrovascular resistance: the brain's own vessels narrowed as blood velocity in its middle cerebral artery, a proxy for its blood flow, fell. The reduced form simplifies this away (main text, Section 3). One candidate explanation is logged under the rule above, not yet checked: lower carbon dioxide from faster breathing narrows the brain's vessels. If confirmed, that would be the governor limiting the top's own access while lower parts still draw, which the model does not allow, so it would count against the model, not explain the finding. It stays uncited until a source has been read and checked. A second candidate is logged under the rule above, not yet checked: as flow falls, the brain can draw a larger share of the oxygen reaching it, so its draw may have held for part of the fall. It will be checked only against a source that measured the brain's oxygen extraction or oxygenation up to presyncope, and the outcome will be logged either way. Until then the finding counts against the model.

Blood loss, mismatch 2: decompensation is an active switch, not a passive failure of control. The model was revised to include a threshold switch, built from this finding.

Fasting (F6): recovery order was consistent for starvation but the opposite under food restriction. Weak, because only a summary was read.

Muscle (M3): performance, taken as the indicator, fell alongside mood in overreaching, so it was not clearly late.

- **Explanation named:** maximal performance tests probe capacity directly, so they are state signals.
- **Outcome:** check 9 tested that rule on five cases, all consistent.

Protection referenced to peak load (FM3): inconsistent as written. A reassignment was made after the result and checked once; it is weak.

Recovery order: the rival rule D (dependency and next demand), named in advance, did not hold as the main driver and was struck in that role.

Weight cycling: two results against are logged; neither has been read beyond its summary:

- no slower loss in the second cycle in female rats (Archambault et al. 1989);
- a rodent study summarised as finding no adverse effect of cycling.

S7.4 A corrected entry, kept for the record

The finding as first logged: in tumour-bearing mice, the brain's glucose use fell while the tumour became the second-largest glucose consumer. It was logged as against brain priority.

The correction: the full abstract states that "brain metabolism in the tumour-bearing state was maintained by an increased use of lactate and 3-hydroxybutyrate" (Mulligan and Tisdale 1991). The brain switched fuel; its energy supply was not cut. For the brain the resource is energy, so the finding is not against brain priority.

The entry is kept to show the correction.

S8. Literature searches

Three searches looked for formal models that set an order of loss among parts under shortage. Each protocol was committed before the search ran. Strings, scripts, raw records, screening sheets (a decision and reason for every record) and counts are in theory/search_check1/, theory/search_check1_s2/ (with theory/search_check1_s2_run1/) and theory/search_check2/. Screening was by title and abstract, by one AI reviewer, except where stated.

S8.1 Check 1, search 1

Protocol: theory/PAM_check1_search_protocol.md, committed before running.

Question: has any published formal model specified the order in which two or more organs, tissues or body parts lose supply under resource shortage, and is that order derived from properties of access?

Databases and strands: PubMed and OpenAlex, all years, run on 7 October 2026. Strand A covered dynamic energy budget models of organs under shortage; strand B covered any formal model of organ priority under shortage.

Classes for includes:

- I-access: order from properties of access;
- I-fixed: order assigned;
- I-emergent: order from another mechanism.

Formal priority among functions (F) and empirical orders with no model (E) were recorded separately.

Counts:

Step	Number
Retrieved	150 records
After deduplication	139
Included	0
Formal function priority (F)	4, all dynamic energy budget models
Empirical order, no model (E)	2

Result: not usable. The search failed its own known-item check. It did not retrieve the one relevant formal model already known (Göbel and colleagues' brain-against-periphery model), and its terms missed plant source-sink allocation. Its null result is reported and not used.

S8.2 Check 1, search 2

Protocol: theory/PAM_check1_search2_protocol.md, committed before running. It was built to close search 1's two gaps:

- **strand C:** plant source-sink allocation models;
- **strand D:** organs competing for energy.

Sensitivity test: set in advance as six known items.

- **Design items (could be used in revision):**
 - K1: Göbel and Langemann 2011 (published online 2010), *Theory Biosci*;
 - K2: Peters and Langemann 2009;
 - K3: Minchin, Thorpe and Farrar 1993;
 - K4: van Leeuwen, Zonneveld and Kooijman 2003.
- **Held out (could not be used in revision):**
 - H1: Thornley 1972;
 - H2: Göbel et al. 2010, *J Theor Biol*.

Revision 1, the only one allowed. Run 1 missed K4, which uses the vocabulary of energetics ("tumour-in-host", "energetics of tumour and host"), not of allocation. A fourth sub-query was added to strand D, and strand C was unchanged. Run 1's records are kept.

Known items, final: K1 to K4 found; H2 found; **H1 missed.**

- The miss is explained by vocabulary alone. H1 is not in PubMed, and neither its title nor its OpenAlex abstract contains any strand C phrase: it speaks of root:shoot ratios, not allocation.
- **Strand C is therefore of limited sensitivity**, as the protocol requires it to be reported.

Deviation D-1 (approved by the author): strand C screened through reviews. Strand C's exhaustive title pass (11,175 titles) was stopped after 410, because the question it was built for was already answered: formal allocation models among plant organs are common, and at least one (K3) derives sink priority from transport properties.

- **Replacement:** a targeted reading of reviews of plant allocation models, chosen by a title rule fixed in the deviation. The reviews are Lacoite 2000, Génard et al. 2008, Franklin et al. 2012, Minchin and Lacoite 2005, and Marcelis and Heuvelink 2007, the last read in full.
- **Result:** four classes of plant model (S3):
 - functional equilibrium;
 - sink regulation by strict priority (hierarchical models);
 - transport-resistance models, in which priority emerges;
 - optimisation models.

Deviation D-2 (a method decision, logged): the OpenAlex title pass of strand D done by AI agents, with a blind check.

- **The task:** 6,809 new titles needed a title pass. Three AI agents in sequence did it under a deliberately liberal rule: keep anything that could be a formal allocation model or a neighbouring theory.

- **The check:** before any agent ran, the reviewer screened a random sample of 300 titles blind (seed 20261007). The agents kept all 10 of the reviewer's sample keeps and marginal cases, plus 12 more.

Counts:

Step	Number
Strand C retrieved	PubMed 1,616; OpenAlex 11,834
Strand D retrieved	PubMed 385; OpenAlex 7,372 (6,809 new)
Final deduplicated set	18,363 records

Includes (strand D, both databases): 21.

- **12 I-access:**
 - the Selfish Brain group's formal models (papers of 2010, 2011 and 2013, listed in the screening sheets; the 2012 review of mathematical models; the 2008 deductive appetite model; the supply-chain models of 2007 to 2011);
 - a tilt model of cerebral perfusion;
 - a brain-centred four-compartment model (2026 preprint);
 - a lifespan brain-health model built on Göbel et al. 2010.
- **7 I-emergent:** dynamic energy budget tumour-in-host models (van Leeuwen et al. 2003, and a 2018 to 2021 series listed in the screening sheets); and a spreading-depression model.
- **1 weak I-emergent:** a GLUT1-deficiency model.
- **1 weak I-fixed:** a 1982 computer model in which an autonomous tumour drain draws first.

Not included:

- 1 provisional I-access (abstract only);
- 5 records still to check (title or abstract only), none likely to change the answer;
- 1 probable exclude;
- 18 near misses (dynamic energy budget models of two organisms: host and symbiont, host and parasite);
- 55 neighbouring theories (to S3);
- 33 empirical records;
- 19 formal models with no allocation under shortage.

Answer to check 1. Formal models with an order of loss among parts exist in three domains: plants (strict priority, or priority emerging from transport), human energy metabolism (the Selfish Brain family, with access gated by insulin), and tumour-host energetics (dynamic energy budget). None has:

- unit loss or scars;
- repair competing for access;
- a ledger of unmet requirement;
- viability;
- use across domains.

S8.3 Check 2: models of the order of organ loss in starvation and haemorrhage

Search: PubMed only, run on 7 October 2026. Five strings, with one revision logged before screening (theory/search_check2/README.md). It returned 1,279 unique records, all titles read, and candidates read as abstracts.

Answer:

- **Haemorrhage: a large formal tradition exists.** It covers the Guyton model and its descendants, and lumped-parameter closed-loop models with baroreflex control (S3). The order among vascular beds is not written in as a rule. It emerges from each bed's resistance, reflex gain and autoregulation, fitted to data: order from properties of access.
- **Fetal brain sparing:** formal circulation models, with the order again set by bed resistances.
- **Starvation across many organs:** no formal model of the order of organ loss was found. The formal models found have two compartments (fat and lean; brain and body). The multi-organ order is held as data, not as a model.

Limits: PubMed only; one reviewer; abstracts only. "Not identified" is not "absent".

S9. Mapping a system, in full

The model makes predictions about a system only once the system is mapped. Every choice below is fixed before any outcome data are opened. Anything chosen with the expected outcome in mind is declared fitted. The corrected H1 mapping (tests/H1 VitalDB G12 - mapping corrected (v2).md), made after H1's result, illustrates the procedure in version 2's terms; it is post hoc, not a test. The setting maps frozen with the two held-out tests remain their test records:

- tests/H1 VitalDB G12 - setting map and mapping (draft).md (its slips are listed in the corrected file);
- tests/PT1 Priority test - setting map and mapping (draft).md.

The order of mapping is fixed: the boundary first, then the top and the protected flow, then everything else. A scenario is a load applied to a system, not a system; never choose the top, the protected flow or an indicator because a data set happens to record it.

S9.1 The nine steps

1. **Boundary, top and protected flow.** Fix the boundary first: it decides everything else, including which part is the top. Then name the top (the part everything else is sacrificed to keep going) and the protected flow (the flow of the resource to the top, against the top's need), and declare the currency. These are the model's main safeguards.
 - **Viable set:** fix it, and fix whether outside support counts as admissible, since that decides collapse against death (S1.11).
 - **Pathways:** the top's work may be an early stage of a longer pathway, which can hold while a shortfall appears in a later, dependent stage. Fix whether viability requires the early output, the downstream completion, or both. The order of failure in what is observed need not be the order of onset.

2. **Resources, carriers and stores.**

- Which resources does work need, and in what ratio?
- Are they complementary (used in fixed proportion)? The law of the minimum applies only where they are. Substitutable inputs are mapped as one resource, or declared (for the brain, glucose and ketones are one resource, energy).
- Which routes do they move along (the network)?
- Which stores exist for each, in what order, with what release profile (proportional, or full until empty) and what turnover?
- Which can be spilled?

3. **The governor and its targets.** Which levels does the governor hold, and with what targets? Which indicators will be watched, and is each the protected flow, a level the governor holds, or neither? What does the governor act through: stores, intake, gates, pathway capacity, explicit budgets? Tissue that makes or carries the governor's signal does work, so it is parts.

4. **Parts.** For each part, fix the following from independent sources:

- its units and capacity;
- its rank for each resource: first its dependency layer (the top, the parts whose work feeds the top, the parts that feed those), then, within a layer, from documented properties of access and, where access is built in, from requirement, or one shared rank if the orders are documented to coincide;
- where access is by saturable uptake, its half-saturation constant, maximum uptake and reference allocation (S9.3);
- its rebuild time, template limit and route back (for fixed capital, none);
- its requirements per unit of work, basal maintenance and renewal (a baseline per active unit plus wear per unit of work);
- its reference allocation, against which shortfall is counted;
- whether it is support for the top, whether it is the intake, and whether it is temporary;
- how economising is realised in it: consolidation (units switched off; the default), or a declared share of units working below capacity.

5. **Repair:** routine repair is part of each part's renewal requirement (step 4). Damage raises the damaged part's requirement (rising requirement), met within its rank. Name any part whose work produces a resource that repair needs (for example, bone marrow), with its rank, as for the intake. Repair cells working at a damaged part are counted at that part, as its units.

6. **Links and pathways:**

- dependencies (supply edges) and loops;
- pathways and their routes (redundancy), and whether flow among routes divides by physics or is placed by access settings (S1.10);
- non-bypassable links (points of severance);
- which parts draw on a shared repair resource.

7. **Nested systems:** only if the proposed extension is used. List every part treated as a nested system, with the evidence for its own access loop, and its interfaces. A part not listed stays an ordinary part.

8. **Modes, gates and requirement links:** only if access-limited shortfall is claimed, or a part's requirement is taken to rise when another part's work falls (S9.2).

9. **The clock:** fix the unit in which durations are judged.

S9.2 The access guard

Shortfall has three origins:

- falling supply;
- rising requirement;
- a governor mode that restricts access while resources suffice.

The third is admitted only for a mode and a gate documented independently, and named at mapping before outcomes.

Requirement links. The same guard applies when rising requirement is traced to another part: a part whose requirement rises when another part's work falls depends on that part's work, and the link counts only if it was documented at mapping, before outcomes. Nothing passes between parts; the requirement link is a dependency. Most of what is called a cascade of work sits inside one part: staff are units of a team, a vacancy is a lost unit, and the team's requirement is unchanged. Work not done that falls on another system, across the boundary, is outside this boundary's ledger and belongs to that system's mapping. Requirement raised by damage follows a rule fixed in advance with the reference allocation, and counts only for damage documented independently. A shortfall cannot be relabelled afterwards as raised requirement.

Modes and snapshots. The top never changes within a boundary. Below it, the order follows from access and requirement. Access settings are fixed within a governor mode and change only when the mode changes. When a sensed level crosses its threshold and the governor acts, the setting is acting, not changing: access changes and the order does not. A mode change is a change in the order, or in which levels the governor holds, other than through built-in access. The model applies to one mode at a time, a snapshot. A threshold switch, such as decompensation, is a mode change: crossing its threshold changes the order, or which levels the governor holds, and ends the snapshot. Modes are documented independently, before outcomes are seen. A gate not named at mapping cannot be invoked later to explain a shortfall. Access cut while the flow is ample, with no documented mode, is the signature of a false alarm (S9.5), named only under that guard. This keeps the access-limited regime of Proposition 1 from becoming a way to explain away any result.

S9.3 Rank under saturable uptake

Where parts take a shared resource by saturable uptake (transporters, binding), document for each part:

- the half-saturation constant K^M ;
- the maximum uptake V ;
- the reference allocation q^0 .

How the rank follows:

- A part loses adequate supply when the pool falls below $C^* = K^M\eta/(1 - \eta)$, with $\eta = q^0/V$.
- **Parts lose adequate supply in descending order of C^* , so the part with the highest C^* ranks lowest and goes short first (S1.7).**
- Affinity alone decides the rank only where η is equal across parts.

- Priority approaches strict where neighbouring thresholds are far apart (S2), and parts go short together where they are close.

S9.4 The bias ledger

Before anything is looked up, each mapping choice is scored for its interpretive freedom, from 1 (forced) to 5 (open). The choices scored are:

- the boundary;
- part classification;
- the resource;
- rank;
- the reference state;
- the prediction.

Coin-flip choices are marked as ambiguous before the search, and cannot be resolved afterwards in the model's favour.

S9.5 Using the model in reverse

Reverse mode starts from what is seen failing. It points to where the shortfall began, which parts' access was being turned down unseen, and which dependency the failing parts share.

- **It suggests where to look; it never names a cause.**
- **Disciplines:**
 - it generates hypotheses and is not evidence;
 - it corrects for what is visible;
 - it names a set, not a cause.
- It must tell failure clustered by rank from failure clustered by dependency (G18).

Signal faults. The propositions assume the governor's signals are true: what it senses matches the real levels, and the access it sets is put into effect; the realised flow may still be limited by the network (S1.9) (main text, Section 3.3). A system with a signal fault is outside the propositions, and for it the model is diagnostic only. Used in reverse, the gap between what a governor with true signals would do and what is observed points to where a signal failed:

- **a false alarm:** access cut while the flow is ample, with no documented mode;
- **a missed alarm:** the order broken, with a higher part short while a lower part still draws, and no part it depends on short in the step before. A documented dependency lag, or a recovery lag (units of the top or a support part still coming back after an earlier shortfall), is not a fault, but the earlier shortfall must be observed, not inferred from the later one;
- **a setting not carried out:** a part's access not matching the setting commanded.

A missed alarm and S1.9. Both describe the top or a support part short while a part outside them is still supplied, with the supports met in the step before and no unit of the top or a support part still coming back. With true signals, S1.9 traces this to a cut or narrowed link, or to outside damage. Reverse mode keeps both in the set and names neither as the cause. A setting not carried out must likewise be told apart from a narrowed route (S1.9).

The guard. A signal fault is never a rescue. It can be named only if it is documented independently, or logged in advance as a candidate and then checked. As above, reverse mode suggests where to look and never names a cause. The access guard (S9.2) and the rule for explanations (S7) apply as well.

S10. Compatible illustrations (not tests)

Cases whose direction fits a proposition but which cannot bear weight. They are known outcomes, read after the model was built.

S10.1 Intensive insulin after ischaemic stroke (Proposition 1, scarcity regime).

- **The trial:** Rosso et al. (2012), INSULINFARCT, *Stroke* 43:2343-2349 (abstract read; closed access). 180 patients with hyperacute ischaemic stroke were randomised to intensive or usual subcutaneous insulin for 24 hours.
- **Results:**
 - intensive insulin gave better glucose control (mean below 7 mmol/L in 95.4% against 67.4%);
 - it went with larger infarct growth, a secondary outcome (median 27.9 against 10.8 cm³; P = 0.04);
 - three-month function was identical (45.6% in both groups);
 - deaths were 10% against 15.6%, not reported as significant.
- **Reading under the model:** correcting a visible figure (blood glucose) without adding resource reopens the insulin-dependent route into muscle and fat. Under the Selfish Brain account, post-stroke hyperglycaemia is the brain's own pull on supply (Sprengell, Kubera and Peters 2021b), so the injured brain would go short instead.
- **Why it is not support:**
 - one trial;
 - a secondary outcome at P = 0.04;
 - no difference in function or death;
 - where the glucose went was not measured;
 - the outcome was known before the model was applied.
- **What a test would need:** a pre-registered study in a system mapped as resource-short, measuring the delivery of the resource to each part before and after a correction outside the named exceptions, with the matching deficit predicted in advance.

S11. Dated record of the model's development

What this is. A dated record of the model's development, with the time stamps of each step. It is not evidence for the model.

- **Sources:** the public repository's commit history, and the author's earlier framework for institutions (dated 27 September 2026).
- **The earlier framework** is held privately and is available from the author on request. Its fingerprint (the SHA-256 of each file) is given in the public repository's README, so a copy supplied on request can be checked against it.
- **Wordings:** "author" is the human author; "AI collaborator" is Claude (Anthropic).
- **Rule:** convergence with earlier work is reported as convergence, never as a successful prediction.

- **Past wording is kept as dated.** "Record" (version 1's term, the ratio of the top's output to its level, which the checks applied to the figure observers watch), "load" (the shortfall), movement language and intentional wording ("on purpose", "demand") appear as they were used at the time. Each row describes the model as it stood on that date; later changes are in the version 2 row.

S11.1 How the model grew

Date (2026)	Step	What changed
By 27 September	The author's earlier framework for institutions	About 60 principles. Candidate formal models: a load ledger, priority queueing and a drift threshold
30 September	First notes on natural systems	Coupling and drift; living systems tie parts' fate to the whole; cancer as a part optimising for itself
3 October	Turn to natural systems	Load conservation stated as a principle; cross-domain cases gathered
4 October	Conserved quantity attempt	Hierarchy of parts, lower parts buffering higher ones, two directions of load, recovery order, part roles (store, intake, working part)
5 October	First queueing form of the model	Parts as servers with stores; load passed down; the break as a store running out
5 October	Twelve natural surface checks, each with predictions committed before sources were opened	Blood loss, fasting, kidney, honeybee colony, plants in drought, fetal growth restriction, rival recovery-order rules, muscle, the record rule (in version 2, the protected flow and indicators), re-tuning, protection referenced to peak load, weight cycling
5 October	Priority as a formula	Order derived from marginal value, not assigned
5 October	Outside reviews absorbed, each logged as a source	Network rather than tree; use in reverse; economising
6 October	Simplifications by the author	Work is local and load is displaced; repair as a store refilling; economising as the governor's action
6 October	The architecture of 6 October	The system dies, not its parts; a separate governor that only allocates; parts with no demand of their own; units as state; repair cut first; basal maintenance before support work
6 October	First held-out test (H1)	Pre-registered and frozen before any data were opened
7 October	Literature read; version 1	Established mathematics imported (DEB forms, viability theory, max-flow min-cut); the formal instances in Section 2 found; propositions proved and checked
8 and 9 October	Version 2	Seven components; the protected flow and indicators; the shortfall; G12's derivation found inconsistent with Proposition 2 and G27 added; the corrected H1 mapping; the laws of the model fixed by the author. Three 6 October choices reversed: the governor is a set of targets on sensed levels, which owns the signals and does no work; the order is rank first, part by part, with upkeep before work only within a part, in place of basal maintenance before support work; and "repair is the first thing cut" became "repair is work like any other": routine repair is part of each part's own draw, and damage raises the damaged part's requirement, met within the part's rank. Repair parts withdrawn; the model describes an arrangement, not an intention. Added on 9 October: signal integrity (the governor's signals assumed true); the order below the top following from access and requirement, with access settings fixed within a mode and a mode change defined; the fourth exception; the dependency lag; rebuilding after a shortfall is not repair

S11.2 Theorised before the matching work was read (class A)

How the work ran. The AI collaborator drafted. The author shaped the model mainly by contesting drafts that did not fit the author's principles or observations. Most class A ideas came from those corrections.

Idea (date fixed in the record)	Whose	Matching work, and when it was found	Agreement and difference
Load is conserved: relocated, not removed (earlier framework; restated 3 October)	Author	Bode integral and conservation of fragility (Csete and Doyle 2002; Doyle and Csete 2011), read 3 October, after the statement. Kleinrock's (1965) conservation law, named 5 October	Same form: priority decides where the shortfall is counted but cannot remove it. Theirs holds across frequencies or within one queue; ours across the parts of a system (Proposition 1)
The governor is a separate function that only allocates; parts have no demand of their own (6 October)	Author, correcting a draft in which the governor set targets	Supply-chain form of the Selfish Brain (Peters and Langemann 2009) and the central governor (Noakes 2012), read 7 October	Same split between a ranking regulator and passive parts. Perceptual control theory (Powers 1973) was noted by the AI collaborator when the idea was recorded. Version 2 reverses this: the governor is a set of targets on sensed levels (S11.1)
Ordered draw: parts drawn in a fixed order from one flow (6 October)	AI collaborator, formalising the author's governor rule	Plant sink priority (Grossman and DeJong 1994; Minchin, Thorpe and Farrar 1993) and the hierarchy of ATP consumers (Buttgereit and Brand 1995), found 7 October	Same ranked order in plants and in the cell; neither has unit loss, scars, a ledger or use in other domains
Repair is the first thing cut when supply falls (6 October)	Author	Triage theory (Ames 2006); Bobba-Alves, Juster and Picard (2022); translational arrest (Hochachka et al. 1996), found 7 October	Same order of cuts in triage theory and translational arrest, as stated then; Bobba-Alves, Juster and Picard state no order among repair, growth and maintenance (S3 note 55). Version 2 withdraws the rule: repair is work like any other, and no rule puts it first or last (S11.1, version 2 row)
Economising: the governor lowers demand and capacity together, on purpose and reversibly (6 October)	Joint	Balanced metabolic suppression (Hochachka et al. 1996), found 7 October	Same reversible, regulated reduction
A tumour acts as a cut in total supply, with the order among the other parts kept (7 October)	Author	DEB tumour-in-host models, found in a screening completed the same afternoon	Same treatment of a tumour as a competing sink. The author had not seen these records
With a store carrying the gap and its release not binding, the break comes at the same cumulative gap whatever the rate (G3, first clause, 5 October)	AI collaborator	Fast and slow haemorrhage in sheep (Scully et al. 2016); fast and slow drought in trees (Dai, Wang and Wan 2018), matched after	Matches the first clause. The AI collaborator knew both in outline beforehand

S11.3 Known when the idea was formed (class B: precedent, no independence claimed)

Idea	Source, and when it was known
Brain spared while organs lose mass in starvation	The Selfish Brain account and Krieger's organ data, logged by the AI collaborator on 3 October. The author did not read it; no claim is made either way

Idea	Source, and when it was known
A hierarchy of parts recruited under stress	Miller's Living Systems Theory, read 4 October
Three states of a part (coping, stressed, compromised)	Selye's general adaptation syndrome, read 4 October (secondary source)
Saturation and sudden decompensation	Woods (2018), read 3 October
Renewal classes of tissues	Bizzozero (textbook pathology)
The point at which a store begins to release	Cherel and Groscolas's fasting phases, read 4 October
The compensatory reserve in blood loss	Convertino's work, known when the blood-loss predictions were written on 5 October
Shared upstream supply and a threshold switch	Built from blood-loss findings after the first natural check, 5 October

S11.4 Imported (class C: cited as the basis)

Taken from existing work; each is cited where it is used, except the newsvendor critical fractile and Michaelis-Menten competition, standard results named without a citation:

- DEB forms (shrinking, synthesising units, proportional release);
- viability theory;
- max-flow min-cut;
- the newsvendor critical fractile;
- metabolic control analysis;
- Michaelis-Menten competition;
- greedy allocation by marginal value.

Elements adopted from outside reviews are logged, with their source, in the author's working records, which are kept privately and are available on request.

S11.5 The caveat that governs class A

- The AI collaborator helped build the model, and its training very likely included the Selfish Brain, DEB, allostasis, Hochachka's and Ames's work, and plant allocation models.
- The author's earlier principles (27 September) predate reading these literatures, but the route as a whole is not independent of them (main text, Section 6), and the model's wording is not either.
- Class A is therefore reported as ideas **theorised beforehand and later found in the literature**, not as predictions and not as evidence.
- The evidence that the ranked order of draws is real is the agreement among the four fields in Section 2, which developed separately.

Supplement references

References cited only in the supplement. Works cited in the main text are listed there and are not repeated here. Each entry was checked against Crossref, PubMed or OpenAlex. How much of each was read is stated where it is used.

- Archambault CM, Czyzewski D, Cordua y Cruz GD, Foreyt JP, Mariotto MJ (1989). Effects of weight cycling in female rats. *Physiol Behav* 46:417-421.

- Aubin J-P (1991). *Viability Theory*. Birkhäuser, Boston.
- Barnes JJ 3rd, Kiberenge K, Sweet R, Keller J, Konia MR (2020). Comparing hemorrhage in human physiology simulation tools: how they compare with expected human physiology and each other. *Simul Healthc* 15:310-317. <https://doi.org/10.1097/SIH.0000000000000427>
- Bizzozero G (1894). An address on the growth and regeneration of the organism. *BMJ* 1:728-732. <https://doi.org/10.1136/bmj.1.1736.728>
- Cherel Y, Groscolas R (1998). Relationships between nutrient storage and nutrient utilisation in long-term fasting birds and mammals. 22nd International Ornithological Congress, Durban, symposium SO1.2 (online proceedings).
- Csete ME, Doyle JC (2002). Reverse engineering of biological complexity. *Science* 295:1664-1669. <https://doi.org/10.1126/science.1069981>
- Curcio L, D'Orsi L, Cibella F, Wagnert-Avraham L, Nachman D, De Gaetano A (2020). A simple cardiovascular model for the study of hemorrhagic shock. *Comput Math Methods Med* 2020:7936895. <https://doi.org/10.1155/2020/7936895>
- Dhabhar FS, McEwen BS (1999). Enhancing versus suppressive effects of stress hormones on skin immune function. *Proc Natl Acad Sci USA* 96:1059-1064. <https://doi.org/10.1073/pnas.96.3.1059>
- Doyle JC, Csete M (2011). Architecture, constraints, and behavior. *Proc Natl Acad Sci USA* 108:15624-15630. <https://doi.org/10.1073/pnas.1103557108>
- Drenos F, Kirkwood TBL (2005). Modelling the disposable soma theory of ageing. *Mech Ageing Dev* 126:99-103. <https://doi.org/10.1016/j.mad.2004.09.026>
- Ford LR, Fulkerson DR (1956). Maximal flow through a network. *Can J Math* 8:399-404. <https://doi.org/10.4153/CJM-1956-045-5>
- Franklin O, Johansson J, Dewar RC, Dieckmann U, McMurtrie RE, Brännström Å, Dybzinski R (2012). Modeling carbon allocation in trees: a search for principles. *Tree Physiol* 32:648-666. <https://doi.org/10.1093/treephys/tp138>
- Garcia-Canadilla P, Rudenick PA, Crispi F, Cruz-Lemini M, Palau G, Camara O, et al. (2014). A computational model of the fetal circulation to quantify blood redistribution in intrauterine growth restriction. *PLoS Comput Biol* 10:e1003667. <https://doi.org/10.1371/journal.pcbi.1003667>
- Génard M, Dauzat J, Franck N, Lescourret F, Moitrier N, Vaast P, Vercambre G (2008). Carbon allocation in fruit trees: from theory to modelling. *Trees* 22:269-282. <https://doi.org/10.1007/s00468-007-0176-5>
- Ghodsi A, Zaharia M, Hindman B, Konwinski A, Shenker S, Stoica I (2011). Dominant resource fairness: fair allocation of multiple resource types. *Proceedings of the 8th USENIX Symposium on Networked Systems Design and Implementation*.
- Göbel B, Langemann D, Oltmanns KM, Chung M (2010). Compact energy metabolism model: brain controlled energy supply. *J Theor Biol* 264:1214-1224. <https://doi.org/10.1016/j.jtbi.2010.02.033>
- Guest H, Munro KJ, Prendergast G, Millman RE, Plack CJ (2018). Impaired speech perception in noise with a normal audiogram: no evidence for cochlear synaptopathy and no relation to lifetime noise exposure. *Hear Res* 364:142-151. <https://doi.org/10.1016/j.heares.2018.03.008>
- Guyton AC, Coleman TG, Granger HJ (1972). Circulation: overall regulation. *Annu Rev Physiol* 34:13-44. <https://doi.org/10.1146/annurev.ph.34.030172.000305>

- Hester RL, Brown AJ, Husband L, Iliescu R, Pruett D, Summers R, Coleman TG (2011). HumMod: a modeling environment for the simulation of integrative human physiology. *Front Physiol* 2:12. <https://doi.org/10.3389/fphys.2011.00012>
- Ibaraki T, Katoh N (1988). *Resource Allocation Problems: Algorithmic Approaches*. MIT Press, Cambridge, MA.
- Kiecolt-Glaser JK, Marucha PT, Malarkey WB, Mercado AM, Glaser R (1995). Slowing of wound healing by psychological stress. *Lancet* 346:1194-1196. [https://doi.org/10.1016/S0140-6736\(95\)2899-5](https://doi.org/10.1016/S0140-6736(95)2899-5)
- Kirkwood TBL (1977). Evolution of ageing. *Nature* 270:301-304. <https://doi.org/10.1038/270301a0>
- Lacoite A (2000). Carbon allocation among tree organs: a review of basic processes and representation in functional-structural tree models. *Ann For Sci* 57:521-533. <https://doi.org/10.1051/forest:2000139>
- Luria O, Bar J, Kovo M, Malinger G, Golan A, Barnea O (2012). The role of blood flow distribution in the regulation of cerebral oxygen availability in fetal growth restriction. *Med Eng Phys* 34:364-369. <https://doi.org/10.1016/j.medengphy.2011.07.032>
- Marucha PT, Kiecolt-Glaser JK, Favagehi M (1998). Mucosal wound healing is impaired by examination stress. *Psychosom Med* 60:362-365. <https://doi.org/10.1097/00006842-199805000-00025>
- Mauritsson K, Jonsson T (2023). A new flexible model for maintenance and feeding expenses that improves description of individual growth in insects. *Sci Rep* 13:16751. <https://doi.org/10.1038/s41598-023-43743-1>
- McCann JC, Ames BN (2009). Vitamin K, an example of triage theory: is micronutrient inadequacy linked to diseases of aging? *Am J Clin Nutr* 90:889-907. <https://doi.org/10.3945/ajcn.2009.27930>
- McCann JC, Ames BN (2011). Adaptive dysfunction of selenoproteins from the perspective of the triage theory: why modest selenium deficiency may increase risk of diseases of aging. *FASEB J* 25:1793-1814. <https://doi.org/10.1096/fj.11-180885>
- Miller JG (1978). *Living Systems*. McGraw-Hill, New York.
- Minchin PEH, Lacoite A (2005). New understanding on phloem physiology and possible consequences for modelling long-distance carbon transport. *New Phytol* 166:771-779. <https://doi.org/10.1111/j.1469-8137.2005.01323.x>
- Mulligan HD, Tisdale MJ (1991). Metabolic substrate utilization by tumour and host tissues in cancer cachexia. *Biochem J* 277:321-326. <https://doi.org/10.1042/bj2770321>
- Nettle D, Andrews C, Bateson M (2017). Food insecurity as a driver of obesity in humans: the insurance hypothesis. *Behav Brain Sci* 40:e105. <https://doi.org/10.1017/S0140525X16000947>
- Noakes TD (2000). Physiological models to understand exercise fatigue and the adaptations that predict or enhance athletic performance. *Scand J Med Sci Sports* 10:123-145. <https://doi.org/10.1034/j.1600-0838.2000.010003123.x>
- Rosso C, Corvol J-C, Pires C, Crozier S, Attal Y, Jacqueminet S, et al. (2012). Intensive versus subcutaneous insulin in patients with hyperacute stroke. *Stroke* 43:2343-2349. <https://doi.org/10.1161/STROKEAHA.112.657122>
- Schulkin J, Sterling P (2019). Allostasis: a brain-centered, predictive mode of physiological regulation. *Trends Neurosci* 42:740-752. <https://doi.org/10.1016/j.tins.2019.07.010>

- Selye H (1936). A syndrome produced by diverse nocuous agents. *Nature* 138:32. <https://doi.org/10.1038/138032a0>
- Shaulson ED, Cohen AA, Picard M (2024). The brain-body energy conservation model of aging. *Nature Aging* 4:1354-1371. <https://doi.org/10.1038/s43587-024-00716-x>
- Sherbrooke CC (1968). METRIC: a multi-echelon technique for recoverable item control. *Oper Res* 16:122-141. <https://doi.org/10.1287/opre.16.1.122>
- Sprengell M, Kubera B, Peters A (2021b). Proximal disruption of brain energy supply raises systemic blood glucose: a systematic review. *Front Neurosci* 15:685031. <https://doi.org/10.3389/fnins.2021.685031>
- Sterling P (2018). Predictive regulation and human design. *eLife* 7:e36133. <https://doi.org/10.7554/eLife.36133>
- Thornley JHM (1972). A balanced quantitative model for root:shoot ratios in vegetative plants. *Ann Bot* 36:431-441. <https://doi.org/10.1093/oxfordjournals.aob.a084602>
- van Leeuwen IMM, Zonneveld C, Kooijman SALM (2003). The embedded tumour: host physiology is important for the evaluation of tumour growth. *Br J Cancer* 89:2254-2263. <https://doi.org/10.1038/sj.bjc.6601394>
- Woods DD (2018). The theory of graceful extensibility: basic rules that govern adaptive systems. *Environ Syst Decis* 38:433-457. <https://doi.org/10.1007/s10669-018-9708-3>